Detailed Syllabus

R22

Metallurgical and Materials Engineering

For

B. Tech. 4-Year Degree Course

(Applicable for the students admitted into E1 from the Academic Year 2022-23)

(I – IV Years Syllabus)

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

Basar, Nirmal, Telangana – 504107

Course Title Page No

Category: Basic Science Course Subject code: MA1101

Linear Algebra and Calculus

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 4- 0- 0- 4

Course Objectives: The students are able to understand

- Matrices and their properties and applying this concept to solve system of linear equations.
- Concept of Eigen values, Eigen vectors and to reduce the quadratic form to canonical form.
- Concept of sequence, nature of series theorems and their application to the mathematical problems.
- Evaluation of surface area, volumes of revolutions, improper integrals using beta and gamma functions.
- Partial differentiation, total derivative and finding maxima & minima of function of variables.

Unit -1: Matrix Theory

Types of matrices, symmetric, Hermitian, skew-symmetry, skew-Hermitian, orthogonal matrices, unitary matrices; elementary row, column operations on a matrix, rank of a matrix by echelon form and normal form, inverse of a non-singular matrix by Gauss-Jordan method; consistency and solutions of system of linear equations using elementary operations, Gauss elimination method; Gauss Seidel iteration method.

Unit -2: Eigen values and Eigen vectors

Linear transformation and orthogonal transformation; characteristic roots and vectors of a matrix; diagonalization of a matrix; Cayley-Hamilton theorem; finding inverse and power of a matrix by Caylay-Hamilton theorem; quadratic forms and nature of the quadratic forms; reductionof quadratic form to canonical form by orthogonal transformation.

Unit -3: Sequences & Series

Definition of a sequence, limit; convergent, divergent and oscillatory sequences. Convergent, divergent and oscillatory series; series of positive terms; comparison test, p-test, D-Alembert's ratio test; Raabe's test; Cauchys Integral test; Cauchys root test; logarithmic test. alternating series; Leibnitz test; alternating convergent series; absolute and conditionally convergence.

Unit -4: Calculus

Mean value theorems: Roll's theorem, Lagrange's mean value theorem, Cauchy's mean value theorem; Taylor's and Macaurin's series with remainders, expansions; applications of definite integrals to evaluate surface area and volumes of revolutions of curves: definition of improper integrals and their convergence, beta and gamma functions and their applications.

Unit -5: Multivariable Calculus

Definitions of limits and continuity. Partial differentiation; Euler's theorem; total derivative; jacobian; functional dependence and independence; maxima and minima of functions of several variablesusing Lagrange multipliers.

Course outcomes: The student will be able to

- Write the matrix representation of set of linear equations and analyze solution of system of equations.
- Find the Eigen values and Eigen vectors and reduce the quadratic form to canonical form.
- Analyze the nature of sequence and series and solve the applications on the mean value theorems.
- Evaluate the improper integrals using beta and gamma functions.
- Find the extreme values of functions of two variables with/without constraints.

Textbooks:

- "Advanced Engineering Mathematics", Erwin Kreyszig, John Wiley and Sons, 10th Edition, 2011.

- "Advanced Engineering Mathematics", R. K. Jain and S. R. K. Iyengar Narosa Publications House, 2008.
- "Higher Engineering Mathematics", B. S. Grewal, Khanna Publications, 2009.

References:

- "Calculus and Analytic Geometry", G. B. Thomas, R. L. Finney, 9^{th} Edition, Pearson, Reprint,2002.
- "Higher Engineering Mathematics", Ramana B. V., Tata McGraw Hill, 11th Reprint, 2010.
- "A Text Book of Engineering Mathematics", N. P. Bail Manish Goyal, Laxmi Publications, 2008.

Category: Basic Science Course Subject code: PH1101

Engineering Physics

Externals: 60Marks L-T-P-C*

Internals: 40Marks 3- 0- 0- 3

Course Objectives:

- To inculcate in the students a sense of yearning to learn the basic physics behind the applications that we look around in day to day life.
- The basic details of solid state physics, optics and electrodynamics and quantum physics provided in a subtle fashion dealt in finer details to have strong basics in these areas.

Unit -1: Vectors and Mathematical Physics

Gradient, divergence, curl and its applications, line, surface and volume integrals, stokes and gauss theorem: applications, curvilinear coordinates: polar, cylindrical and spherical co-ordinates, problems.

Unit -2: Electrodynamics

Electrodynamics before Maxwell, fixing of ampere's law, Maxwell equations in matter, boundary conditions, continuity equation, Poynting theorem, wave equation for E and B, monochromatic plane waves, energy and momentum in EM waves, propagation in linear media, reflection and transmission at normal incidence. EM waves in conductors, reflection at conducting surface.

Unit -3: Quantum Mechanics

Introduction to quantum mechanics, De-Broglie's waves and uncertainty principle, significance of wave Function, time dependent Schrodinger wave equation, time independent Schrodinger wave equation, particle in a box – problems.

Unit -4: Electron Structure of Solids

Introduction to crystallography, Bravais lattices and crystal systems, atomic packing, atomic radii, crystal structures (SC, BCC and FCC), Miller indices, classical free electron theory, Kronig Penny model (e vs k), band theory of solids.

Unit -5: Semiconductor Physics

Intrinsic and extrinsic semiconductors, Fermi level and carrier-concentration, effect of temperature on Fermi level. Mobility of charge carriers and effect of temperature on mobility, Hall effect, energy band gap determination of semiconductors by four probe method, direct and indirect band gap semiconductors.

Course outcomes: The students will be able to

- understand the innate physics principles that go into the day to day phenomenon specially to optical domain.
- get hold of the basic electromagnetic wave concepts that are crucial in understanding the communication phenomenon.
- realize the difference between the Newtonian domain (classical physics) and quantum domain (quantum mechanics) and get to know the physics that happens at the quantum domain.
- get equipped with concepts in understanding the crystals and materials and know the physics behind the properties exhibited by these materials.

Textbooks:

- "Engineering Mechanics", MK Harbola, Cengage Learning India, 2nd Edition, Jan 2013
- "An Introduction to Mechanics", D Kleppner & R Kolenkow, 2nd Edition, Tata Mc Graw Hill Publisher, July 2017
- "Solid state Physics", S. O. Pillai, NEW AGE International,6th edition
- "Principles of Engineering Physics", Khan and Panigrahi, Cambridge publishers, March 2017

References:

- "Engineering Physics", Malik and Singh
- "Electrodynamics", David.J.Griffiths
- "Quantum Mechanics", Aruldhas
- "Solid State Physics", C. Kittel

Category: Humanities and Social Sciences course Subject code: HS1101

English

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 2- 0- 0- 2

Course Objectives:

- Improve the language proficiency of students in English with an emphasis on vocabulary, grammar,reading and writing skills.
- Equip students to study academic subjects more effectively and critically using the theoreticaland practical components of English syllabus.
- Develop study skills and communication skills in formal and informal situations.

Unit -1: The Raman Effect

Vocabulary building: the concept of word formation, the use of prefixes and suffixes grammar: identifying common errors in writing with reference to articles and prepositions. Reading: reading and its importance techniques for effective reading. Basic writing skills: sentence structures use of phrases and clauses in sentences importance ofproper punctuation techniques for writing precisely paragraph writing: types, structures and features of a paragraph - creating coherence-organizing principles of paragraphs in documents.

Unit -2: Ancient Architecture in India

Vocabulary: synonyms and antonyms. Grammar: identifying common errors in writing with reference to noun-pronoun agreement and subject-verb agreement. Reading: improving comprehension skills techniques for good comprehension writing: format of a formal letter-writing formal letters e.g., letter of complaint, letter of requisition,job application with resume.

Unit -3: Blue Jeans

Vocabulary: acquaintance with prefixes and suffixes from foreign languages in English to form derivatives- words from foreign languages and their use in English. Grammar: identifying common errors in writing with reference to misplaced modifiers and tenses. Reading: sub-skills of reading- skimming and scanning writing: nature and style of sensible writing- defining- describing objects, places and events classifying-providing examples or evidence

Unit -4: What Should You be Eating

Vocabulary: standard abbreviations in English grammar: redundancies and clichés in oral and written communication. Reading: comprehension- intensive reading and extensive reading writing: writing practices–writing introduction and conclusion - essay writing-precis writing.

Unit -5: How a Chinese Billionaire Built her Fortune

Vocabulary: technical vocabulary and their usage grammar: common errors in English reading: reading comprehension-exercises for practice writing: technical reports- introduction characteristics of a report categories of reports formats-structure of reports (manuscript format) -types of reports - writing a report.

Course outcomes: The students will be able to

- use English language effectively in spoken and written forms.
- comprehend the given texts and respond appropriately.
- communicate confidently in various contexts.
- acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

Textbooks:

- “English for Engineers”, Sudarshana, N.P. and Savitha, C. Cambridge University Press, (2018).

References:

- “Practical English Usage”, Swan, M, Oxford University Press, (2016)

- “Communication Skills”, Kumar, S and Lata, P, Oxford University Press, (2018).
- “Remedial English Grammar”, Wood, F.T. Macmillan, (2007).
- “On Writing Well Zinsser”, William, Harper Resource Book, (2001).
- “Study Writing”, Hamp-Lyons, L Cambridge University Press, (2006).
- “Exercises in Spoken English. Parts I III”. CIEFL, Hyderabad. Oxford University Press.

Category: Program Core Course Subject code: MM1101

Introduction to Materials Engineering

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- To get a broad overview on various aspects of Materials classification and crystal structures.
- To understand the mechanical properties of materials
- To get an idea of defects in materials
- To get awareness of physical and magnetic properties of materials
- To know the heat treatment processes of metals and alloys

Unit-1: Classification of material

Classification of metals, ceramics and polymers in terms properties, applications. Crystal structures: co-ordination number, effective number atoms, correlation between a lattice parameter and radius, atomic packing factor; Miller Indices: indexing of lattice planes and directions; density calculations: linear and planar density.

Unit-2: Mechanical behavior and processing

Mechanical properties of materials, stress-strain diagram, types of mechanical loads, plastic deformation of metals, Introduction to mechanical working processes of metals: rolling, forgin g, extrusion, pilgering, wire drawing, tube drawing

Unit-3: Materials defects

Point defects: Vacancy, interstitial and substitutional defects, line defects: edge dislocations and screw dislocations, area defects: grain boundary, surface, volume defects: void, precipitate

Unit-4: Properties of materials

Electrical properties: electrical conductors and insulators, semi-conductors: extrinsic and intrinsic, direct and indirect semi-conductors, super conductors: type-1, type-2 and high temperature super conductors; thermal properties: Thermal conductivity, heat capacity and thermal expansion; magnetic properties: hysteresis loop, soft and hard magnetic materials, and its application in material science.

Unit-5: Heat treatment processes for metals

Heat treatment of steels, scheme of heat treatment process, improvement of mechanical properties by heat treatment, annealing process, normalizing process, hardening process and tempering process, scheme of heat treatment process

Course Outcomes: The student will be able to

- Get awareness on materials classification and crystal structures.
- Understand the mechanical properties of materials
- Understand the materials defects
- Get awareness on physical and magnetic properties of materials
- Know the heat treatment processes of metals and alloys

Text books:

- "Physical Metallurgy Principles", R. E. Reed Hill, 1991.
- "Physical Metallurgy: Principles And practices", V. Raghavan, 2006

References:

- "Material Science and Engineering, William Callister , Wiley, 1985.
- "Physical metallurgy principles", Reza Abbaschian, Cengage; 2013
- "Fundamentals of Physical Metallurgy", John D. Verhoeven, 2000
- "Physical Metallurgy", Robert W. Cahn and Peter Haasen, Vol. I, II and III, 2000

Category: Engineering Science Course Subject code:EE1101

Basic Electrical and Electronics Engineering

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3- 0- 0- 3

Course Objectives: This course introduces

- Electrical dc and ac circuits, basic law's of electricity and methods to solve the electrical net works
- Construction operational features of energy conversion devices: transformers, dc, induction motors.
- Basics of electronics, semiconductor devices and their characteristics and operational features.

Unit -1: Circuits Analysis

Electrical circuit elements: R-L-C Parameters, V-I relationship for passive elements, diode, voltage andcurrent Independent and dependent sources, circuit analysis: Kirchoff's Laws, network reduction techniques series, parallel, series parallel, star-to-delta, delta-to-star transformation, source Transformation, mesh Analysis and nodal analysis network theorems – Thevenin's, Norton's maximum power transfer, superposition, step response of RL, RC and RLC circuits

Unit -2: AC Circuit Analysis

Single Phase AC circuits – R.M.S. and average values, form factor, steady state analysis of series, parallel and series parallel combinations of R, L and C with sinusoidal excitation, concept of reactance, impedance, susceptance and admittance phase and phase difference, concept of power factor, j-notation, complex and polar forms of representation. Resonance series resonance and parallel resonance circuits

Unit -3: Three Phase AC Circuits

Three phase EMF generation, delta and Y connections, line and phase quantities, solution of three phase circuits, balanced supply voltage and balanced load, phasor diagram, measurement of power in three phase circuits

Unit -4: Basic Electronics

Introduction to electronics and electronic systems, diode and rectifier circuits (Half and Full wave), BJT, transistor biasing. Small signal transistor amplifiers (CE), operational amplifiers and their basic application, introduction to digital circuits

Unit -5: Electrical Machines

Transformers: construction, EMF equation, ratings, phasor diagram on no load and full load, equivalent circuit, regulation and efficiency calculations, open and short circuit test, applications, DC machines: construction, EMF and torque equations, characteristics of DC generators and motors, applications, induction motors: the revolving magnetic field, principle of orientation, ratings, equivalent circuit,Torque-speed characteristics, applications

Course outcomes: The student will be able to

- Understand the basic concept of electrical circuits under dc and ac excitation
- Understand basic concept and performance of transformers and motors used as various industrial drives

Textbooks:

- "Electrical Technology", Hughes Prentice Hall, 7th edition
- "Millman's Electronic Devices and Circuits", J.Millman and C.C.Halkias, Satyabratajit, TMH, 2/e, 1998.
- "Electric Machines", I.J. Nagrath & D.P. Kothari, Tata McGraw Hill, 7th edition, 2005

References:

- "Electronic Devices and Circuits", K. Lal Kishore, B.S. Publications, 2nd Edition, 2005.

- "Electronic Devices and Circuits", Anil K. Maini, VarshaAgarwal Wiley India Pvt. Ltd., 2009.
- "Network Theory", N.C. Jagan & C. Lakshminarayana, B.S. Publications.
- "Network Theory", Sudhakar, Shyam Mohan Palli, TMH.

Category: Humanities and Social Sciences course Lab Subject code: HS1701

English Lab

Externals: 60 Marks L-T-P-C*

Internals: 40 Marks 0- 0- 2- 1

Course Objectives:

- To facilitate computer-assisted multi-media enabling individualized independent language learning.
- To sensitize students to the nuances of English speech sounds, word accent, intonation and rhythm.
- To bring about a consistent accent and intelligibility in students' pronunciation.
- To improve the fluency of students in spoken English
- To train students to use language appropriately for public speaking and interviews

Unit -1: Listening Skills

Listening skill- its importance purpose process types barriers of listening communication at work place spoken vs. Written language. Practice: introduction to phonetics speech sounds vowels and consonants ice breaking activity and jam session situational dialogues greetings taking leave introducing oneself and others.

Unit -2: Structure of Syllables

Structure of syllables word stress and rhythm weak forms and strong forms in context- features of good conversation non-verbal communication. Practice: basic rules of word accent - stress shift - weak forms and strong forms in context-situational dialogues role-play- expressions in various situations making requests and seeking permissions - telephone etiquette.

Unit -3: Intonation-Errors in Pronunciation

Intonation-errors in pronunciation-the influence of mother tongue (mti)- how to make formal presentations. Practice: common Indian variants in pronunciation differences in British and American pronunciation- formal presentations.

Unit -4: Public Speaking Exposure

Listening for general details-public speaking exposure to structured talks. Practice: listening comprehension tests- making a short speech extempore

Unit -5: Interview Skills

Listening for specific details- interview skills practice: listening comprehension tests- mock interviews.

Course outcomes: The students will be able to

- Better understanding of nuances of english language through audio- visual experience and groupactivities
- Neutralization of accent for intelligibility
- Speaking skills with clarity and confidence which in turn enhances their employability skills

References:

- Connected Speech- Software
- Issues in English 2- Software
- http://www.clarityenglish.com/program/practicalwriting/
- http://www.clarityenglish.com/program/roadtoielts/
- http://www.clarityenglish.com/program/clearpronunciation1/
- http://www.clarityenglish.com/program/resultsmanager/

Category: Basic Science Course Lab Subject code: PH1701

Engineering Physics Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0- 0- 3- 1.5

Course Objectives:

- To make the students gain practical knowledge to co-relate with the theoretical concepts.
- To achieve perfectness in experimental skills and the study of practical applications will bring more confidence and ability to develop and fabricate engineering and technical equipments.
- Design of circuits using new technology and latest components and to develop practical applications of engineering materials and use of principle in the right way to implement the modern technology.

The list of experiments are:

1. Photoelectric effect
2. Hall effect
3. Ultrasonic interferometer
4. Melde's experiment
5. Four probe method
6. Frank hertz experiment
7. Seebeck and Peltier effect
8. Solar cell
9. Coupled pendulum

Course outcomes: Students will be able to

- Develop skills to impart practical knowledge in real time solution.
- Understand principle, concept, working and application of new technology and comparison of results with theoretical calculations.
- Design new instruments with practical knowledge
- Gain knowledge of new concept in the solution of practical oriented problems and to understand more deep knowledge about the solution to theoretical problems.
- Understand measurement technology, usage of new instruments and real time applications in engineering studies.

References:

- "Engineering Mechanics", MK Harbola, 2nd Edition, Cambridge Publishers.
- "An Introduction to Mechanics", D Kleppner & R Kolenkow, 2nd Edition, Tata Mc Graw Hill Publishers
- NPTEL Lecture Notes by MK Harbola
- "Solid State Physics", S.O.Pillai, NEW AGE International
- "Engineering Physics", Dr. M. N. Avadhanulu, Dr. P. G. Kshirsagar, S.Chand Publications
- "Solid State Physics and Electronics", R. K. Puri and V. K. Bubbar, S.Chand Publications
- "Materials Science and Engineering", V. Raghavan, Eastern Economy Edition
- "Principles of Engineering Physics", Khan and Panigrahi, Cambridge publishers

Category: Engineering Science Course Subject code: CE1701

Engineering Graphics

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0- 1- 4- 3

Course Objectives:

- To introduce the "Universal Language of Engineers" for effective communication through drawing.
- To understand the basic concepts of drawing through modern techniques.
- To impart knowledge about standard principles of projection of objects.
- To provide the visual aspects of Engineering drawing using Auto-CAD.

Unit -1: Introduction to Engineering Drawing

Introduction to engineering drawing: principles of engineering graphics and their significance, usage of drawing instruments, lettering, types of lines and dimensioning. Overview of auto-cad: the menu system, tool bars, drawing area, dialogue boxes, short cut menu, the command lines, select and erase objects, introduction to layers etc., drawing simple figures-lines, planes, solids.

Unit -2: Geometrical Constructions

Geometrical constructions: construction of regular polygons. Conic sections: construction of ellipse, parabola, hyperbola (general method only), cycloid, epicycloids, hypocycloid and involutes. Scales: construction of plain, diagonal and vernier scales.

Unit -3: Orthographic Projections

Orthographic projections: principles of orthographic projections. Projections of points: projections of points placed in different quadrants. Projection of lines: lines parallel and inclined to both the planes (determination of true lengths and true inclinations and traces). Projection of planes: planes inclinedto both the reference planes

Unit -4: Projection of Solids

Projection of solids: projection of solids whose axis is parallel to one of the reference planes and inclined to the other plane, axis inclined to both the planes projection of sectioned solids: sectioning of simple solids like prism, pyramid, cylinder and cone in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other obtaining true shape of section.

Unit -5: Development of Surfaces

Development of surfaces: development of surfaces of right regular solids-prism, pyramid, cylinder and cone isometric projections: principles of isometric projection isometric scale, isometric views of planes andsimple solids perspective projections: basic concepts of perspective views.

Course outcomes: The student will be able to

- Use engineering principles and techniques to understand and interpret engineering drawings.
- Understand the concepts of auto-cad.
- Draw orthographic projections of lines, planes and solids using auto-cad.
- Use the techniques, skills and modern engineering tools necessary for engineering practices.

Textbooks:

- "Engineering Drawing", Bhat N.D., Panchal V.M. & Ingle P.R., Charotar Publishing House, (2014).
- "EngineeringDrawing" Gopala krishna K.R., (Vol.I & II combined), Subhas Stores, Bangalore, 2007.

- "Engineering Drawing and Computer Graphics", Shah, M.B. & Rana B.C, Pearson Education, (2008).

References:

- "EngineeringGraphics", Venugopal K. and Prabhu Raja V., New Age publications
- "Engineering Graphics", Agrawal B. & Agrawal C.M, TMH Publication, (2012).
- "Text book on Engineering Drawing", Narayana, K.L. & P Kannaiah, Scitech Publishers, (2008).
- Auto CAD Software Theory and User Manuals

Category: Engineering Science Course Lab Subject code:EE1701

Basic Electrical and Electronics Engineering Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0- 0- 2- 1

Course Objectives:

- To expose the students to the concepts of electrical and electronics circuits and their applications
- To expose the students to the operation of dc machines and transformer and give them experimental skills.

List of Laboratory Experiments:

1. Open circuit & short circuit test on single phase transformer.
2. Verification of KCL & KVL.
3. Characteristic of the lamps (Tungsten, Fluorescent and Compact Fluorescent Lamps)
4. Verification of network theorems.
5. V-I characteristics of Diodes and BJT
6. Half-wave and full-wave rectifiers, rectification with capacitive filters, zener diode
7. Studies on logic gates

Course outcomes: Student will be able to

- Understand principles of measuring instruments of voltage, current and power
- Analyze the characteristics of semiconductor devices and understand their applications
- Analyze the characteristics and evaluate performance of dc machines and transformers

Introduction to lab:

- Basic safety precautions. Introduction and use of measuring instruments voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors and inductors.
- Demonstration of cut-out sections of machines: dc motor (commutator-brush arrangement), induction motor (squirrel cage rotor), synchronous motor (field winging - slip ring arrangement) and single-phase induction motor.
- Demonstration of Components of LT switchgear.
- Measuring the steady-state and transient time-response of R-L, R-C, and R-L-C circuits to a step change in voltage (transient may be observed on a storage oscilloscope). Sinusoidal steady state response of R-L, and R-C circuits impedance calculation and verification. Observation of phase differences between current and voltage. Resonance in R-L-C circuits.
- Transformers: Observation of the no-load current waveform on an oscilloscope (non sinusoidal wave-shape due to B-H curve nonlinearity should be shown along with a discussion about har- monics). Loading of a transformer: measurement of primary and secondary voltages and currents, and power.

Textbooks:

- “Basic Electrical Engineering”, D. P. Kothari and I. J. Nagrath, Tata McGraw Hill, 2010.
- “Basic Electrical Engineering”, D. C. Kulshreshtha, McGraw Hill, 2009.
- “Fundamentals of Electrical Engineering”, L. S. Bobrow, Oxford University Press, 2011.
- “Electrical and Electronics Technology”, E. Hughes, , Pearson, 2010.
- “Electric Machines”, I.J.Nagrath & D.P.Kothari, Tata Mc Graw Hill, 7th Edition.2005

References:

- “Electrical Engineering Fundamentals”, V. D. Toro, Prentice Hall India, 1989.
- “Engineering Circuit Analysis”, William Hayt and Jack E. Kemmerly, Mc Graw Hill Company, 6th edition.

- "Network Theory", N.C.Jagan & C.Lakshminarayana, B.S. Publications.
- "Network Theory", Sudhakar, Shyam Mohan Palli, TMH.
- "Electrical Machines", PS Bhimbra, Khanna Publishers.

Category: Basic Science Course Subject code: MA1201

Differential Equations and Vector Calculus

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 4- 0- 0- 4

Course Objectives:

- To study the Methods of solving the differential equations of first and higher order.
- To study the methods of solving improper integrals and the concepts of multiple integrals
- The basic properties of vector valued functions and their applications to line, surface and volume integral.
- To study numerical methods to analyze an experimental data.

Unit -1: Ordinary Differential Equations -1

Ordinary differential equations of first order: exact first order differential equation, finding integrating factors, linear differential equations, Bernoulli's, Riccati, Clairautss differential equations, finding orthogonal trajectory of family of curves, Newtons Law of cooling, law of natural growth or decay.

Unit -2: Ordinary Differential Equations-2

Ordinary differential equations of higher order: linear dependence and independence of functions, Wronskian of n- functions to determine linear independence and dependence of functions, solutions of second and higher order differential equations (homogeneous & non-homogeneous) with constant coefficients, method of variation of parameters, Euler-Cauchy equation.

Unit -3: Integral Calculus

Integral calculus: convergence of improper integrals, tests of convergence, Beta and Gamma functions - elementary properties, differentiation under integral sign, differentiation of integrals with variable limits - Leibnitz rule. Rectification, double and triple integrals, computations of surface and volumes, change of variables in double integrals - Jacobians of transformations, integrals dependent on parameters applications.

Unit -4: Vector Differentiation

Vector differentiation: vector point functions and scalar point functions. Gradient, divergence and curl. Directional derivatives, tangent plane and normal line. Vector Identities. Scalar potential functions. Solenoidal and irrotational vectors.

Unit -5: Vector Integration

Vector integration: line, surface and volume integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

Course outcomes: The student will be able to

- Solve first order linear differential equations and special nonlinear order equations like bernouli, riccati & clairauts equations
- Compute double integrals over rectangles and type i and ii regions in the plane
- Explain the concept of a vector field and make sketches of simple vector fields in the plane.
- Explain concept of a conservative vector field, state and apply theorems that give necessary and sufficient conditions for when a vector field is conservative, and describe applications to physics.
- Recognize the statements of stokes' theorem and the divergence theorem and understand howthey are generalizations of the fundamental theorem of calculus.
- Able to solve the problems in diverse fields in engineering science using numerical methods.

Textbooks:

- "Advanced Engineering Mathematics", R. K. Jain and S. R. K. Iyengar, 3rd Edition, Narosa Publishing House, New Delhi

References:

- "Advanced Engineering Mathematics", Erwin Kreyszig, 8th Edition, Wiley-India.
- "Ordinary and Partial differential equations", Dr. M.D. Rai singh ania, 17th Edition , S.CHAND, 2014.

Category: Basic Science Course Subject code:CY1201

Engineering Chemistry

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3- 0- 0- 3

Course Objectives:

- To understand the importance of the spectroscopy in determining the structures of chemicalcompounds
- To understand the importance of electrochemistry in technical field
- To understand the rates of some of the reactions and derivation of their rate laws
- To understand the phase rule with some examples
- To understand the importance of materials in the technical field

Unit -1: Electrochemistry

Introduction to electrochemistry: Galvanic cell (Daniel cell), Nernst equation. Types of electrodes: metal-metal ion electrodes, metal-insoluble salt-anion electrodes, calomel electrode, gas-ion electrodes, hydrogen and chlorine electrodes, oxidation-reduction electrodes (quinhydrone electrode), amalgam electrodes and ion exchange electrode (glass electrode). EMF and applications of EMF: determination of pH of the solution, potentiometric titrations, determination of the valency of the ions, solubility product of sparingly soluble salts. Thermodynamic data: enthalpy and entropy of cell reactions, Gibbs- Helmholtz equation and applications. Classification of commercial cells - primary cells (dry cell) and secondary cells (Lithium-ion battery, Pb-acid storage battery). Fuel cells: $H_2 – O_2$ fuel cell, methanol- oxygen fuel cell, phosphoric acid fuel cell.

Unit -2: Corrosion and Water Treatment

Dry and wet corrosion and their mechanisms. Pilling - Bedworth Rule. Types of corrosion: galvanic corrosion, concentration cell corrosion, pitting corrosion and stress corrosion. Factors influencing therate of corrosion: Temperature, pH and dissolved oxygen. Corrosion Prevention methods: cathodic protection sacrificial anodic method and impressed current method. Metallic coatings: galvanization and tinning methods. Water: Hardness of water, degrees of hardness. Calculation of hardness by EDTA method. Disadvantages of hard water in boilers: priming, foaming, scales, sludges and caustic embrittlement. Treatmentof boiler feed water: Zeolite process, Ion exchange process.

Unit -3: Energy Sources

Introduction. Definition and classification of fuels. Calorific value of a fuel, Characteristics of a good fuel. Coal, types of coal. Analysis of coal: proximate and ultimate analysis. Bomb calorimeter and Junkers gas calorimeter. Problems on calculation of calorific value. Liquid fuels petroleum Extraction fractional distillation. Synthetic Petrol: Bergius process and Fisher Tropsch process. Bio-fuels: bio- diesel, bio-gas.

Unit -4: Chemical Kinetics

Introduction to rate of reaction and rate constant determination. Factors influencing rate of reaction. Complex reactions: definition and classification of complex reactions, definition of reversible reactionswith examples, rate law derivation for reversible reactions. Consecutive reactions: definition, rate law derivation and examples of consecutive reactions. Parallel reactions: definition, rate law derivation and examples of parallel reactions. Steady-state approximation: introduction, kinetic rate law derivation by applying steady state approximation in case of the oxidation of NO and pyrolysis of methane.

Unit -5: Nanochemistry

Introduction to nanomaterials, classification: Carbon based nanomaterials, metallic nanoparticles, metal oxide nanoparticles. Properties at nanoscale. Synthetic approaches: Top-Down (Lithography, spray pyrolysis, FIB, ball milling) and bottom-up (Sol-gel, Hydrothermal, CVD, PVD). Brief overview on characterization of nanomaterials: UV, X-ray, SEM and TEM. Applications of nanomaterials.

Course outcomes: At the end of the course student will be able to

- solve first order linear differential equations and special non linear first order equations like bernouli , riccati & clairauts equations
- compute double integrals over rectangles and type i and ii regions in the plane
- explain the concept of a vector field and make sketches of simple vector fields in the plane.
- explain concept of a conservative vector field, state and apply theorems that give necessary and sufficient conditions for when a vector field is conservative, and describe applications to physics.
- recognize the statements of stokes' theorem and the divergence theorem and understand how they are generalizations of the fundamental theorem of calculus.
- solve the problems in diverse fields in engineering science using numerical methods.

References:

- “Engineering Chemistry”, Jain & Jain
- “Engineering Chemistry”, Shashi Chawla
- “Chemistry for Engineers”, B. K. Ambasta
- “Engineering Chemistry”, H. C. Srivastava
- “Fundamentals of Engineering Chemistry”, Shikha Agarwal

Category: Engineering Science Course Subject code:CS1201

Programming for Problem Solving

Externals: 60Marks L-T-P-C*
Internals: 40Marks 3- 0- 0- 3

Course Objectives:

- To introduce the basic concepts of computing environment, number systems and flowchart.
- To familiarize the basic constructs of C language data types, operators and expressions
- To understand modular and structured programming constructs in C
- To learn the usage of structured data types and memory management using pointers
- To learn the concepts of data handling using pointers

Unit-1: Arithmetic expressions

Introduction to programming & arithmetic expressions and precedence: introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.). Idea of algorithm: steps to solve logical and numerical problems. Representation of algorithm: flowchart/pseudocode with examples. From algorithms to programs: source code, variables (with data types) variables and memory locations, syntax and logical errors in compilation, object and executable code- arithmetic expressions and precedence

Unit-2: Arrays

Conditional branching, loops & arrays: writing and evaluation of conditionals and consequent branching, iteration and loops arrays (1-D,2-D), character arrays and strings

Unit-3: Functions

Function & basic algorithms: functions (including using built in libraries), parameter passing in functions, call by value, passing arrays to functions: idea of call by reference searching, basic sorting algorithms (bubble, insertion and selection), finding roots of equations, notion of order of complexity through example programs (no formal definition required)

Unit-4: Structure

Recursion & structure: recursion, as a different way of solving problems. Example programs, such as finding factorial, Fibonacci series, Ackerman function etc. Quick sort or merge sort. Structures, defining structures and array of structures

Unit-5: Pointers

Pointers & file handling: idea of pointers, defining pointers, use of pointers in self-referential structures, notion of linked list(no implementation) file handling (only if time is available, otherwise should be done as part of the lab)

Course outcomes: The students will be able to

- Formulate simple algorithms for arithmetic and logical problems.
- test and execute the programs and correct syntax and logical errors.
- implement conditional branching, iteration and recursion.
- decompose a problem into functions and synthesize a complete program.
- use arrays, pointers and structures to formulate algorithms and programs.

Textbooks:

- "Schaum's Outline of Programming with C", Byron Gottfried, McGraw-Hill, 2017
- "Programming in ANSI C", E. Balaguruswamy, Tata McGraw-Hill 8th edition, 2019

References:

- "The C Programming Language", Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall ofIndia, 1978.

Category: Mandatory Course Subject code:BM1201

Constitution of India

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3- 0- 0- 0

Course Objectives:

- Understand the formation and principles of Indian constitution.
- Understand fundamental rights and its implications in life
- Understand fundamental duties of individual toward country and society
- Understand directive principles to govern the policy formation
- Understand the way of running the government and basic governance

Unit -1: Introduction to Indian Constitution

Meaning of the term constitution, historical background of Indian constitution, making of Indian constitution, constituent assembly

Unit -2: Features of Indian Constitution

Preamble of the constitution, importance, scope, relevance, the salient features of indian constitution, importance, scope, relevance

Unit -3: Fundamental Rights

Fundamental rights, importance and scope of fundamental rights, categorization of fundamental rights

Unit -4: Fundamental Duties & The Directive Principles of State Policy

Fundamental duties, importance and scope of fundamental duties, the directive principles of state policy, importance, scope, relevance

Unit -5: Union/Central Government

Union government, union legislature (parliament), lok sabha and rajya sabha (with powers and functions), union executive, president of India (with powers and functions), prime minister of India (with powers and functions)

Course outcomes: The students will be able to

- Understand the formation and principles of indian constitution.
- Understand fundamental rights and its implications in life
- Understand fundamental duties of individual toward country and society
- Understand directive principles to govern the policy formation
- Understand the way of running the government and basic governance

Textbooks:

- “Indian Polity”, Laxmikanth
- “Indian Administration”, Subhash Kashyap
- “Indian Constitution”, D.D. Basu
- “Indian Administration”, Avasti and Avasti

Category: Program Core Course Subject code:MM1201

Physics of Materials

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3- 0- 0- 3

Course Objectives: To understand

- the classical free electron theory
- quantum mechanical improvements to the free electron model
- incorporating crystal structure into the model
- addressing specific material properties using the models developed

Unit -1: Relationship between thermal and electronic conductivity

Introduction and approach, properties of materials and some important relationships, free electron theory of metals, Drude model electronic conductivity, Drude model thermal conductivity - ratio the Wiedemann Franz law

Unit -2: Maxwell Boltzmann & Fermi Dirac statistics

Maxwell Boltzmann statistics, limitations of the Drude model, elementary quantum mechanics: history and significant concepts, the Drude Sommerfield model, Fermi Dirac statistics, density of states, Fermi energy and Fermi surface, improvements over Drude model, remaining limitations.

Unit -3: Diffraction Condition and Its Significance for Electron Energy

Specific heat, phonons, real space Vs reciprocal space, diffraction condition and its significance for electron energy, Wigner Seitz cells, Brillouin zones, band theory, density of occupied states, the origin of anisotropy

Unit -4: Properties of Materials

Electrons and holes, classification of semiconductors, direct band gap, indirect band gap, opto-electronic materials, magnetic properties, superconductivity, Meissner effect.

Unit -5: Physics of Nanoscale Materials

Bose-Einstein Statistics, BCS theory, high temperature superconductors, physics of nano scale materials

Course outcomes: The student will be able to

- Describe the mechanical, electrical, thermal and optical properties of materials.
- Analyze the importance of material properties for a wide variety of engineering situations.
- Understand the micro-physics and chemistry responsible for material properties, and analyze its modification.
- Evaluate and select suitable materials for different practical applications.
- Get glimpses of typical range for properties of common materials.

Textbooks:

- Prathap Haridoss, “Physics of Materials Essential Concepts of Solid-State Physics”, Wiley 2015.
- David Jiles, “Introduction to Electronic Properties of Materials”, Chapman and Hall, 1994.
- Ashcroft and Mermin, “Solid State Physics”, Saunders College Publishing, 1976.

References:

- David Jiles, “Introduction To Electronic Properties of Materials”, Chapman and Hall, 1994
- John Wulf R.M. Rose and L.A. Shepard, “Structure and Properties of Materials - Electronic Properties”, Wiley Eastern, 1964
- Alan Cottrell, “Introduction to the Modern Theory of Metals”, Ashgate Publishing Company, 1988
- Laszlo Solymar and D. Walsh, “The Electrical Properties of Materials”, Oxford Univ. Press, 1988.

Category: Basic Science Course Subject code:CY1801

Engineering Chemistry Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0- 0- 3-1.5

Course Objectives:

- To learn the preparation of organic compounds in the laboratory.
- To estimate the hardness and alkalinity of the given sample of water.
- To understand the Job's method for determining the composition.
- Learn how to use the pH meter and Polarimeter.

The list of experiments are:

1. Determination of the strength of weak acid (CH_3COOH) by pH metry.
2. Conductometric titration (strong acid (HCl) vs strong base (NaOH)).
3. Throwing power of Copper.
4. Estimation of alkalinity of water.
5. Determination of total hardness of water by complexometric method using EDTA.
6. Determination of the calorific value of fuel sample by using bomb calorimeter.
7. Preparation of bio-diesel from palm oil by trans esterification method.
8. The rate constant and order of the reaction of the hydrolysis of an ester catalyzed by an acid (dil.HCl).
9. Preparation of Nano particle (ZnO).

Course outcomes:
The students get the knowledge on basic synthesis, quantitative and qualitative analysis is being important.

Reference books:

1) Essentials of experimental engineering chemistry by Shashi chawla.
2) Practical chemistry by Dr.O.P.Pandey, S.Chand publication.
3) A textbook of engineering chemistry by Shashi chawla.
4) College practical chemistry by VK Ahluwalia.
5) Practical engineering chemistry by K. Mukkanti.
6) Laboratory manual by R. Kulakarni, Adil.

Category: Engineering Science Course Subject code:ME1802

Engineering Workshop

Externals: 60Marks L-T-P-C*
Internals: 40Marks 0- 1- 2-2

Course Objectives:

- To understand the basic manufacturing process of producing a component by casting, forming plastic molding, joining processes, machining of a component either by conventional or by unconventional processes.
- To understand the advanced manufacturing process of additive manufacturing process.

List of Experiments:

1. **Fitting –** Step and V Fit
2. **Carpentry** – Half lap joint and Dove tail joint
3. **House Wiring** – Series, Parallel, Staircase and Godown wiring
4. **Tin Smithy** – Tray and Cylinder
5. **Welding** – Bead formation, Butt and Lap joint welding
6. **Foundry** – Mold preparation with Single piece and Split piece pattern
7. **Machining** – Plain turning, Facing, Step and Taper turning
8. **Plastic molding** – Demo
9. **WIRE EDM, CNC, 3D Printer** - Demo

Course Outcome:

- Students will gain knowledge of the different manufacturing processes which are commonly employed in the industry, to fabricate components using different materials.

Text Books:

(i) Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Vol. I 2008 and Vol. II 2010, Media promoters and publishers private limited, Mumbai.

(ii) Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", 4th edition, Pearson Education India Edition, 2002.

Reference Books

(i)Gowri P. Hariharan and A. Suresh Babu,"Manufacturing Technology – I" Pearson Education, 2008.

(ii) Roy A. Lindberg, "Processes and Materials of Manufacture", 4th edition, Prentice Hall India, 1998.

(iii) Rao P.N., "Manufacturing Technology", Vol. I and Vol. II, Tata McGrawHill House, 2017.

Category: Basic science Course Subject code:MA2101

Probability and Laplace Transforms

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3- 0- 0- 3

Course Objectives:

- To solve the differential & integral equations using Laplace Transform.
- To understand the applications of Laplace Transforms.
- To understand the concepts of random variables, expectation, variance.
- To learn various distributions and their applications.
- To know the basic concepts of statistics applicable in estimation and testing.

Unit-1: Random Variables

Definition of random variables. Properties of discrete and continuous random variable. Definition and properties of probability mass function and probability density function. Definition of cumulative distribution function and its properties for discrete and continuous distributions.

Sums of independent random variables; expectation of discrete random variables, variance of sum of random variables, correlation coefficient, Chebyshev's Inequality.

Discrete Distributions: Properties of various discrete distributions: binomial, Poisson, distributions, negative binomial and discrete uniform distributions.

Unit-2: Continuous Distributions

Properties of various continuous distributions: uniform, exponential, normal, gamma distributions.

Bivariate Distributions: Definition and properties of bivariate distribution (continuous and discrete). Joint probability distributions. Marginal probability distributions. Conditional probability distributions.

Distributions of sums and quotients of random variables.

Unit-3: Basic Statistics

Measures of Central tendency: Moments, skewness and Kurtosis - Probability distributions: Binomial, Poisson and Normal - evaluation of statistical parameters for these three distributions, Correlation and regression – Rank correlation

Unit-4: Laplace Transform

Definition of Laplace Transform, linearity property, conditions for existence of Laplace Transform. First and second shifting properties, Laplace Transform of derivatives and integrals, unit step functions, Dirac delta-function, error function. Differentiation and integration of transforms, convolution theorem,

Unit-5: Inverse Laplace Transform

Finding inverse Laplace transform using various methods, evaluation of integrals by Laplace transform. Solving initial and boundary value problems, differential equations & partial differential equations, integral equations using Laplace transforms.

Course outcomes: After learning the concepts of this paper the student must be able to

- use shift theorems to compute the Laplace transform and inverse Laplace transform
- use the Laplace transform to compute solutions of equations involving impulse functions
- compute conditional probabilities directly and using Bayes' theorem, and check for independence of events.
- set up and work with discrete random variables. In particular, understand the binomial Poisson Negative Binomial and uniform distributions
- work with continuous random variables. In particular, know the properties of uniform, normal exponential and gamma distributions.

TEXTBOOKS:

1. Sheldon Ross, A First Course in Probability, 6th Ed., Pearson Education India, 2002.
2. Gupta, S.C., Kapoor V.K., Fundamentals of Mathematical Statistics (11th Edition), Sultan Chand & Sons, 2002.
3. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed.,Wiley, 1968.
4. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 8th Edition,
5. R.K.Jain and S.R.K.Iyengar Advanced Engineering Mathematics, Narosa Publications House.2008

REFERENCES:

1. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11thReprint, 2010.
2. B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 2009.
3. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal
   - Book Stall, 2003 (Reprint).

Category: Program Core Course Subject code: MM2101

Physical Metallurgy

Externals: 60 Marks L-T-P-C*
Internals: 40Marks 3-1-0-4

Course Objectives:

- To know the significant roles of microstructure features on mechanical properties.
- To describes generic classification of materials and their specific engineering applications.
- To obtain knowledge about the physical metallurgy of specific and a few important engineering materials.
- To learn about the principles of alloy design, phase diagram and strengthening mechanisms in different metals and alloys.

Unit -1: Introduction

Structure of metals, nucleation kinetics: homogeneous nucleation, rate of homogeneous nucleation, growth rate equation, heterogeneous nucleation, solidification of pure metal and an alloy, concept of super cooling.

Unit -2: Diffusion in Solids

Diffusion mechanisms, Fick's laws of diffusion, Steady state and non-steady state diffusion, solution to Fick's second law, diffusion calculations, Kirkendall effect, Darken's equation, atomic theory of diffusion, Factors that influence diffusion, industrial applications of diffusion processes.

Unit -3: Phase Diagrams

Experimental methods for construction of equilibrium diagrams, unary phase diagram, isomorphous systems, phase rule and its applications, lever rule and its applications, equilibrium heating and cooling of an isomorphous alloy, binary phase diagrams, eutectic systems, congruent melting intermediate phases, eutectoid, peritectic, peritectoid, monotectic and syntectic reactions. Examples of binary systems of $Fe\text{-}Fe_3C$, Cu-Zn, Cu-Sn, Al-Cu and Al-Si.

Unit -4: Iron and Iron Carbide Diagram

$Fe\text{-}Fe_3C$ phase diagram; Effect of alloying elements on $Fe\text{-}Fe_3C$ phase diagram; isothermal and continuous cooling transformation diagrams (TTT and CCT diagrams); influence of alloying elements on transformation characteristics.

Unit -5: Importance of Ferrous and Nonferrous Alloys

Introduction to important ferrous alloys, stainless steels, special steels, cast irons, aluminium alloys.

Course outcomes: The student will be able to

- Understand the solidification mechanisms.
- Explain the significance of Fe-Fe3C diagram.
- Understand the correlation between structure and properties of materials.

Textbooks:

- W. D. Callister, "Materials Science and Engineering: An Introduction", Wiley, New York (2003).
- V. Raghavan, "Materials Science and Engineering", Prentice Hall India, New Delhi, (1998).
- S. N. Avner, "Introduction to Physical Metallurgy", Tata McGraw Hill, 1997.
- Peter Haasen, "Physical Metallurgy", Cambridge University Press, 1996.
- R. E. Reed-Hill and R. Abbaschian, "Physical Metallurgy Principles", 3rd edition, PWS-Kent Publishing, 1992.
- R. E. Smallman, "Modern Physical Metallurgy", Butterworths, 1963.

Category: Program Core Course Subject code:MM2102

Metallurgical Thermodynamics

Externals: 60 Mark L-T-P-C*

Internals: 40 Marks 3-1-0-4

Course Objectives:

- To understand the basics of thermodynamics
- To understand the concept of entropy and significance of auxiliary functions
- To understand the concepts of chemical equilibrium and related concepts
- To understand the concepts of solutions and its relation between thermodynamics and phase equilibrium
- To understand the concepts of electro metallurgy and their relation with thermodynamics

Unit -1: Introduction

Importance of thermodynamics, definition of thermodynamic terms; concept of states, systems equilibrium. Equation of states, extensive and intensive properties, homogeneous and heterogeneous systems. Internal energy, heat capacity, Cp-Cv relations, enthalpy, isothermal and adiabatic processes.

Unit -2: The Second Law of Thermodynamics and Auxiliary Functions

The second law of thermodynamics, entropy, degree of reversibility and irreversibility, criteria of equilibrium. Statistical interpretational of entropy.

Auxiliary functions, combined statements, free energy, Maxwell's relations, transformation formula, Gibbs- Helmholtz equation.

Unit -3: The Third Law of Thermodynamics

Concept of Third law, consequences of third law. Temperature dependence of enthalpy and entropy, Debye and Einstein concept of heat capacity, fugacity, equilibrium constant, Ellingham diagrams, E-pH diagram, thermodynamics of point defects, chemical potential, use of Potential-functions, controlled atmospheres, homogeneous and heterogeneous equilibria.

Unit -4: Thermodynamics of Solutions and Phase Equilibrium

Solution thermodynamics, Solutions, partial molar quantities, ideal and non-ideal solutions, Henry's law, Gibbs - Duhem equation, activity, regular solution, quasi-chemical approach to solution, statistical treatment. Change of standard state. Phase relations and phase rule- its applications. Free energy composition diagrams for binary alloy systems.

Unit -5: Eletcro Metallurgy and Kinetics

Thermodynamics of electrochemical cells, solid electrolytes. Thermodynamic applications in extraction, refining of metals and materials processing.

Kinetics of homogeneous reaction: chemical reaction rate, first order reaction, higher order reactions, determination of reaction order and rate constant, temperature dependence of reaction rate, Heterogeneous reactions.

Course outcomes: Students will be able to

- Understand the concept of state of the system and thermodynamic equilibrium in materials
- Understand the concept of entropy in materials and significance of Gibbs energy and Helmholtz energy through the natural variables of various auxiliary functions
- Calculate the enthalpy and entropy changes for the given system, understand the reactions involving gaseous components
- Understand modeling of solution phases and their application to phase equilibrium in binary systems
- Understand the concepts of electrowinning and electro refining and thermodynamic associated with them

Textbooks:

- D. E. L. David R. Gaskell, *Introduction to the Thermodynamics of Materials,* 6th ed., CRC Press, 2017.
- G. S. UPADHYAYA and R. K. DUBE, *Problems in Metallurgical Thermodynamics and Kinetics*. Elsevier, 1977.
- R. A. Swalin, *Thermodynamics of Solids*. Wiley, 1962.
- L. S. Darken, R. W. Gurry, and M. B. Bever, *Physical Chemistry of Metals*. McGraw-Hill, 1953.
- A. Ghosh, *Textbook of Materials and Metallurgical Thermodynamics*. PHI Learning, 2002.
- K. K. Prasad, H. S. Ray, and K. P. Abraham, *Chemical and Metallurgical Thermodynamics*. New Age International (P) Limited, 2006.

Category: Program Core Course | Subject code:MM2103

Mechanics of Solids

Externals: 60 Marks | L-T-P-C*
Internals: 40 Marks | 3-1-0-4

Course Objectives:

- To enable engineering students analyze basic mechanics problems and apply vector based approach to solve them.
- To understand resolution of forces under equilibrium conditions for rigid bodies in two and three dimensions.
- To understand basic concepts of stress, strain and their relations for elastic and plastic deformation.
- To understand the basic concepts stress transformations, principle stresses and Mohr's circle of stresses in two and three dimensions.
- Introduce the concept of theories of elastic failure and their significance in designing a component.

Unit -1: Introduction to system of forces

Resultant of two forces, resolution of a force into components, unit vectors, equilibrium of a particle, free-body diagrams, vector product of two vectors, moment of a force about a point, rectangular components of the moment of a force, scalar product of two vectors, mixed triple product of three vectors, moment of a force about a given axis.

Unit -2: Equilibrium of Rigid Bodies

Equilibrium of rigid bodies: reactions at supports and connections for a two- and three-dimensional structure, equilibrium of a rigid body in two and three dimensions.

Unit -3: Stresses and Strains.

Elastic and plastic behaviour, engineering stress and strain, tensile deformation of ductile material, ductile vs. brittle behaviour, concept of stress and types of stresses, concept of strain and types of strains. Flow curve, true stress and true strain. True versus engineering strain, advantages of true strain over engineering strain, Poisson's ratio and volume strain, relationship between true and engineering stresses during plastic deformation, invariants of stress and strain

Unit -4: Stress and Strain Relationships for Elastic behaviour

Description of stress at a point, state of stress in two dimensions (plane stress), Mohr's circle of stress two dimensions (plane stress), state of stress in three dimensions, stress tensor, Mohr's circle three dimensions, description of strain at a point, Mohr's circle of strain, hydrostatic and deviator components of stress, elastic stress-strain relations.

Unit -5: Elements of the Theory of Plasticity

Yielding criteria for ductile metals, Failure theories – Maximum principal stress theory, Maximum principal strain theory, Strain energy and shear strain energy theory and the yield locus.

Course outcomes: The student should be able to

- determine resultant of forces acting on a body and analyse equilibrium of a body subjected to a system of forces.
- draw free-body diagrams and write the appropriate equilibrium equations to solve engineering problems dealing with various forces.
- evaluate the stresses, strains and correlate to deformation in various materials.
- understand the stress transformations and its graphical representation through Mohr's circle.
- understand the different criteria for the safety of the component by applying the theories of failure.

Textbooks:

- Ferdinand P. Beer, E. Russell Johnston Jr., David Mazurek, Phillip J. Cornwell, Brian Self, "Vector Mechanics for Engineers: Statics and Dynamics, Tenth Edition", McGraw-Hill, 2012.
- Russell C. Hibbeler, "Engineering Mechanics: Statics, 14th edition", Pearson, 2020.
- George E. Dieter, "Mechanical metallurgy (SI metric edition)", McGraw-Hill, 1988.

References:

- Ferdinand L. Singer, "Engineering Mechanics" Collins, 1975.
- Khurmi R.S, Khurmi N, "Engineering Mechanics", S. Chand, 2020.
- Amit Bhaduri, "Mechanical Properties and Working of Metals and Alloys", Springer Series in Material Science, Volume 264, 2018.
- Barry J. Goodno and James M. Gere, "Mechanics of Materials" Ninth Edition, Cengage Learning, 201

Category: Program Core Course Subject code:MM2104

Non Metallic Materials

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives: To understand the structure, properties, processing and performance of,

- Polymers
- Ceramics
- Glasses
- Carbonaceous materials
- Composites

Unit -1: Polymers

Classification and applications of non-metallic materials, understanding on polymer structures, classification of polymers: thermosetting, thermoplastic, defects in polymers, mechanical behavior of polymers, characteristics of polymers and advanced polymeric materials, processing of polymers vulcanization, polymer additives, forming techniques for plastics and elastomers.

Unit -2: Ceramics

Understanding on ceramic structures, classification of ceramics, traditional ceramics advanced ceramics, refractories, defects in ceramics, phase equilibrium, mechanical behavior of ceramics, thermal, magnetic, dielectric and superconducting behavior of ceramics, characteristics of ceramics and advanced ceramic materials, processing of ceramics

Unit -3: Glasses

Understanding on glass structures, classification of glasses, major composition of glasses, glass former, characteristics of glass and advanced glass materials annealed glass, toughened glass and processing of glass, preparation, melting, forming and finishing. Glass fibers glass tubing

Unit -4: Carbonaceous Materials

Understanding on carbonaceous materials structures, defects in carbonaceous materials, classification of carbonaceous materials, carbon nanotubes, graphite, graphene, CNT, fullerenes, characteristics of carbonaceous Materials, processing of carbonaceous materials CVD, laser ablation, arc discharge

Unit -5: Composite Materials

Understanding on composite materials structures, classification of composite materials, characteristics of composite materials, processing of composite materials, particle - reinforced composites, and fiber reinforced composites, metal matrix composites, ceramic matrix composites, structural composite preparation of ceramic powders: auto combustion, sol- gel synthesis, microwave assisted hydrothermal synthesis

Course outcomes: The student will be able to

- understand technological, economic aspects and engineering application of polymers.

- understand technological, economic aspects and engineering application of ceramics.
- understand technological, economic aspects and engineering application of glasses.
- understand technological, economic aspects and engineering application of carbonaceous materials.
- understand technological, economic aspects and engineering application of composites.

Textbooks:

- Fred W. Billmeyer, Textbook of Polymer Science, Wiley, 2007
- Kingery, Bowen and Uhlman, Introduction to Ceramics, Wiley India Pvt Limited, 2012
- Krishan K. Chawla, Composite Materials: Science and Engineering, Springer, 2012.

References:

- Nieves López-Salas and Markus Antonietti, Carbonaceous Materials: The Beauty of Simplicity, Bulletin of the Chemical Society of Japan, Vol.94, No.12, 2021
- Michio Inagaki, Feiyu Kang, in Materials Science and Engineering of Carbon, 2016
- Ben T Atban, Introduction to Non-metallic materials, Kindle Unlimited, 2017
- G. Hartwig, History and Applications of Nonmetallic Materials, springer, 2007.

Category: Program Core Course Lab Subject code: MM2701

Physical Metallurgy Lab

Externals: 60 Marks L-T-P-C*
Internals: 40Marks 0-0-3-1.5

Course Objectives:

- To provide hands on experience in preparation of sample for metallographic examination
- To study and understand microstructures of various metals.
- To analyze microstructure features from metallographic tests and analysis.

List of experiments

1) Demonstration of metallurgical optical microscope.
2) Metallographic specimen preparation, sample cutting, mounting, and grinding.
3) Comparative study of Microstructures of Low, Medium & High carbon steels and variation of hardness.
4) To study the Microstructure alloyed steels samples (stainless steels or duplex stainless steels)
5) To study the Microstructure Non – Ferrous pure metals. (Copper or Aluminum).
6) To study the Microstructure Non-Ferrous alloys. (Brass or Bronze).
7) To study the Microstructure of eutectic alloys Al-Si, Pb-Sn, and Pb-Sb
8) Microstructure of Cast Iron (Gray, White, Nodular).
9) Qualitative and quantitative analysis (grain size measurement and use of Image J software)
10) Recovery, Recrystallisation and Grain growth of cold worked copper
11) Macro etching and sulphur printing.
12) To study the Electro polishing of given metallic samples (Stainless steels and aluminium alloys.

Course outcomes: The student will be able to

- Understand the working principle of optical microscope.
- Prepare ferrous and non-ferrous metals and alloys sample preparation by following standard metallographic procedures.
- Identifying the phases present in given micrograph.
- Measure the grain size and quantification of phases in given micrograph.

Category: Humanities and Social Sciences course Subject code: BM2201

Managerial Economics and Financial Analysis

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-0-0-3

Course Objectives: To understand,

- the nature and scope of managerial economics and the concepts of demand analysis.
- the significance of demand elasticity and the concepts of demand forecasting.
- the concepts of production and cost analysis.
- the concepts of production and cost analysis different market structures and their competitive situations.
- the concept and significance of capital budgeting.

Unit-1: Introduction to managerial economics
Definition, nature and scope, basic economic principles, the concept of opportunity cost, marginalism, incremental concept, time perspective, discounting principle, risk and uncertainty.

Unit-2: Theory of demand
Demand, demand function, law of demand, determinants of demand and types of demand, elasticity of demand and types, demand forecasting, need for demand forecasting, methods of demand forecasting. Supply: law of supply.

Unit-3: Theory of production
Production meaning, production function, production function with one variable, production function with two variables: isoquants and isocosts, marginal rate of technical substitution, returns to scale. Cost concepts: meaning of costs, types of costs.

Unit-4: Market structure
Classification of market structures, features, competitive situations.
Pricing practices: price output determination under perfect competition, oligopoly, monopoly, features of monopolistic competition; pricing strategies.

Unit-5: Capital
Introduction, definition of capital, sources of capital.
Capital budgeting : significance, need for capital budgeting decisions, capital budgeting decisions, kinds of capital budgeting decisions, methods of capital budgeting - traditional methods (payback period and accounting rate of return methods), discounted cash ow methods, net present value method.

Course outcomes: Students will be able to understand

- economic principles in business.
- forecast demand and supply.
- production and cost estimates.
- market structure and pricing practices.
- economic policies.

Textbooks:

- “BusinessEconomics", H L Ahuja, S.Chand & Co, 13e, 2016.
- “Business Economics", Chaturvedi, International Book House, 2012.
- “Managerial Economics", Craig H. Petersen, W. Cris Lewis and Sudhir K. Jain, Pearson, 14e,2014.
- “Managerial Economics", Dominick Salvatore, Oxford Publications, 7e, 2012.
- ”Business Environment", Justin Paul, Tata McGraw Hill, 2010.
- “Business Environment Text & Cases", Francis Cherunilam, Himalaya Publications, 2012.

Category: Humanities and Social Sciences course Subject code: HS2201

Essence of Indian Traditional Knowledge

Externals: 60 Marks	L-T-P-C*
Internals: 40 Marks	3-0-0-3

Unit-1: Basic Structure of Indian Knowledge System

Veda Definition Kinds Upavedas (Ayurveda, Gandhra veda, Shilpa veda, Artha veda)- Vedangas (Shik-

sha, Kalapa, Chhanda, Niruktha, Vyakarana, Jyothishya), Dharma Shastra, Mimansa, Purana, Tarka Shastra

Unit-2: Modern Science

Modern Science and Indian Knowledge System

Yoga Holistic Health Care

Unit-3: Indian Philosophical Tradition

A) Orthodox School: Samkya, Yoga, Nyaya, Vaisheshika, Purva Mimansa, Vedantha

B) Heterodox School: Jainism, Buddhism, Ajivika, Anjana, Charvaka

Unit-4: Linguistic Tradition

Indian Linguistic Tradition

Unit-5: Indian Artistic Tradition

Chithra Kala (Painting), Sangeetha Kala (Music), Nruthya Kala (Dance)

Category: Program Core Course Subject code: MM2201

Mechanical Properties of Materials

Externals: 60 Marks	L-T-P-C*
Internals: 40 Marks	3-1-0-4

Course Objectives:

- To study various mechanisms of dislocations and their interactions.
- To study various strengthening mechanisms in materials.
- To have a basic knowledge of various mechanical tests like tension, compression, hardness and impact.
- To study deformation by fracture.
- To study deformation by fatigue and creep

Unit -1: Dislocation theory

Imperfections in solids, deformation by slip, slip in a perfect lattice, slip by dislocation movement, critical resolved shear stress, edge, screw and mixed dislocation, burger vector, stress field around dislocation, dislocation glide and climb, forces on dislocations, forces between dislocations, dislocation and plastic strain, dissociation of dislocations, dislocation multiplication, deformation by twinning, jogs and kinks, stacking faults, dislocation pileups.

Unit -2: strengthening mechanisms

Types of strengthening mechanisms, grain boundary strengthening, hall-petch relation, hall-petch strengthening limit, strengthening from second phase, factors influencing second-phase particle strengthening, solid solutions strengthening, precipitation hardening, precipitation sequence-gp zones, factors affecting precipitation hardening, interaction between particles and dislocations-particle cutting and orowan mechanism, coherent and incoherent precipitates, fibre strengthening, martensitic strengthening, ausforming process, strain hardening or cold working, annealing of cold-worked metal-recovery, recrystallisation and grain growth.

Unit -3: Mechanical Testing

Tension Test: introduction to tension test and tensile properties, conditions for necking, effect of temperature and strain rate on tensile properties. Compression: elastic and plastic range, baushinger effect, buckling, barreling. Bending and torsion test. Hardness: classification of hardness, Moh’s scale, brinell hardness, rockwell hardness, vicker’s hard-ness, micro-hardness. Impact test: notched bar impact test and its significance, charpy and izod tests, dbtt curve and its importance, metallurgical factors affecting the transition temperature, temper embrittlement.

Unit -4: Fracture

Types of fracture in metals, theoretical cohesive strength, stress concentration factor, effect of notch, griffith theory of brittle fracture, elastic strain energy release rate, stress intensity factor, fracture toughness.

Unit -5: Fatigue, Creep and stress rupture

Fatigue failure, stress cycles, S-N curve, fatigue crack nucleation and growth, effect of mean stress, stress concentration effect, surface effects and surface treatments, effect of metallurgical variables, temperature effect.

The high-temperature materials problem, creep curve, stress-rupture test, strain-time relationship, creep rate- stress-temperature relationship, creep deformation mechanism.

Course outcomes: The student will be able to

- Understand the concept of plastic deformation and the role of dislocations in plastic deformation of materials
- Understand various strengthening mechanisms to enhance mechanical properties of metals.
- Understand different mechanical testing procedures and calculate mechanical properties like yield strength, hardness and toughness.
- Distinguish between different deformation mechanisms
- Understand the factors influencing deformation and measures to prevent it.

Textbooks:

- George E. Dieter, “Mechanical metallurgy (SI metric edition)”, McGraw-Hill, 1988.
- Amit Bhaduri, “Mechanical Properties and Working of Metals and Alloys”, Springer Series in Material Science, Volume 264, 2018.
- Derek Hull and D.J. Bacon, “Introduction to Dislocations”, Pergamon Press, 2008.

References:

- ASM Metals Hand Book, “Failure Analysis and Prevention”, Vol. 11, 10th Edition, ASM International, 2002.
- Richard W. Hertzberg, “Deformation and Fracture Mechanics of Engineering Materials”, 5th Edition, John Wiley & Sons, New York, 2012.
- David Broek, “Elementary Engineering Fracture Mechanics”, Martinus Nijhoff Publishers, Hingham, Massachusetts, 3rd Edition, 1984.
- William F. Hosford, “Mechanical Behavior of Materials”, Cambridge University Press 2005

Category: Program Core Course | Subject code:MM22

Metal Casting and Joining

Externals: 60 Marks | L-T-P-C*

Internals: 40 Marks | 3-1-0-4

Course Objectives:

- To study and understand the basic concepts of casting
- To study and understand the basic concepts of various casting operations
- To study and understand the basic concepts of Metal Joining
- To study and understand various welding processes
- To study and understand the metallurgy of weldment , testing for welding defects

Unit -1: Introduction to casting

Introduction and classification of casting techniques, melting, solidification, sand casting: tools of sandcasting, moulding sands, moulding sand properties and testing, core and core sands, pattern colors, pattern materials, pattern allowances, types of patterns, making of moulding sand, design of riser andgating system, sand casting defects.

Unit -2: Casting processes and operations

Permanent mould and special casting process: die casting, investment casting, vacuum sealed mouldingand squeeze casting, centrifugal casting, evaporative pattern casting and plaster moulding, continuous casting, finishing and inspection: shakeout, fettling and finishing, testing and quality control.

Unit -3: Introduction to Joining processes

Introduction, applications, classification, Soldering, Brazing, Mechanical Joining, Welding, Welding positions, Weld joints, arc welding processes, arc characteristics, shielded metal arc welding (SMAW), features of SMAW, V-I characteristics, electrodes used in SMAW.

Unit -4: Welding processes

Metal inert gas welding (MIG), submerged arc welding (SAW), gas metal arc welding (GMAW), electro slag welding (ESW), Electro gas welding (EGW), tungsten inert-gas welding (TIG), plasma arc welding (PAW); Resistance welding advantages, applications. Thermo chemical welding and atomic hydrogen welding; laser beam and electron beam welding; solid state welding diffusion, ultrasonic, explosive, friction and forge welding; gas welding; oxy acetylene welding, types of oxy acetylene welding;

Unit -5: Weld ment Tests and Analysis

Weldability, Welding Metallurgy, Welding defects, Inspection and Testing of weldments, Destructive and Non Destructive Testing.

Course outcomes: The student will be able to

- Explain the basic concepts of casting
- Describe various casting operations
- Explain basic concepts of Metal Joining
- Describe various welding processes
- Describe the metallurgy of weldment and defects associated with it

Textbooks:

- R. W. Heine, C. R. Loper, P. C. Rosenthal, ”Principles of metal casting”, Mc Graw Higher Ed,1976.
- P. K. Jain, “Principles of foundry technology”, Mc Graw-Hill 1987.
- K. Easterling, “Introduction to physical metallurgy of welding”, Butterworth-Hienemann 1992.
- Sindo Kou “Welding Metallurgy”, John Wiley & sons Publications 2003

Reference:

- ASM Handbook Volume 6: Welding, Brazing, and Soldering, ASM International, 1993.
- ASM Handbook Volume 15: Casting, ASM International, 2008

Category: Program Core Course Subject code:MM2203

Phase Transformations

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- Correlate thermodynamics and phase stability
- To understand the principles of solidification.
- To analyze the mechanisms and phenomenon associated with diffusion.
- To study and understand pearlitic and bainitic transformations.
- To study and understand the diffusionless transformations

Unit-1 Introduction

Phase equilibrium: Introduction, thermodynamics and stability of phases, classification of phase trans- formations, order of transformation, Gibbs rule and application, phase diagrams, construction and interpretation.

Unit-2 Liquid-Solid Transformation

Nucleation: homogeneous and heterogeneous, growth: continuous and lateral, interface stability; alloy solidification: cellular and dendritic, eutectic, off-eutectic, peritectic solidification, welding, casting andrapid solidification.

Unit-3 Diffusion

Atomic mechanism, interstitial and substitutional diffusion, atomic mobility, tracer diffusion in binary alloys and diffusion in multiphase binary systems. Solid state diffusive transformation: classification, nucleation and growth - homogeneous and heterogeneous mechanism, precipitate growth under different conditions, age hardening, spinodal decomposition, precipitate coarsening, transformation with start range diffusion, recrystallization, grain growth, eutectoid transformation, discontinuous reactions.

Unit-4 Pearlitic And Bainitic Transformation

Factors influencing pearlitic transformation, mechanism of transformation, nucleation and growth, orientation relationship, degenerate pearlite. Bainite: mechanism of transformation, nucleation and growth, orientation relationships, surface relief, classical and non-classical morphology, effect of alloy-ing elements.

Unit-5 Non-Diffusive Transformation

Characteristics of transformation, thermodynamics and kinetics, nucleation and growth, morphology, crystallography, stabilization, strengthening mechanisms, non-ferrous martensite, shape memory ef-fect/alloys and glass transition concept.

Course outcomes:

The student should be able to

- Explain the basics of thermodynamics and the concept of phase stability
- Explain the principles of solidification and use them to control the microstructure during solidification
- To analyze the mechanisms and phenomenon associated with diffusion.
- Describe the pearlitic and bainitic transformations in steels.
- Describe the diffusionless transformations in steels and nonferrous alloys

Textbooks:

- Porter, D, A And Easterling, K. E., "Phase Transformations In Metals And Alloys", 2 nd Edition, CRC Press, 1992.
- Reed-Hill, R, E And Abbaschian, R., "Physical Metallurgy Principles", 3rd Edition, PWS-Kent Publishing Company, 1994.

Reference books:

- Ahindra Ghosh, "Textbook of Materials and Metallurgical Thermodynamics", Prentice Hall of India Pvt. Ltd., 2003
- David R. Gaskell and David E. Laughlin, "Introduction to the Thermodynamics of Materials", CRC Press, Taylor and Francis Group, 2017, 6th Edition.
- Paul G. Shewmon, "Transformations in Metals", Indo American Books.

Category: Program Core Course Subject code: MM2204

Iron and Steel Making

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- To study history of iron production and various methods of producing iron from its ores.
- To study the physical chemistry of iron production
- To study and understand sponge iron production .
- To study and understand thermodynamics and kinetics of the reactions during steel making
- To understand continuous casting processes and materials balance in iron and steel making

Unit -1: Raw materials of Iron making

History of Iron; Occurrence of iron ore, limestone and coke in India; Raw materials for blast furnace pig iron production; Coke production, properties of coke, recoverable and non- recoverable coke ovens process; Agglomeration of Iron ore fines: Sintering, Dwight-Lloyd (DL) sintering, fluidized bed sintering; Pelletisation: disc and drum palletization processes.

Unit -2: Blast furnace design and production

Blast furnace (B/F) profile and design considerations; Physical chemistry of Iron making; Furnace zones: combustion zone, RAFT zone, cohesive zone, thermal reserve zone, chemically inactive zone, B/F refractory lining, gas cleaning system, B/F gas storage stoves.

Unit -3: Alternative Iron making processes

Blast furnace Blow-in process, blow-out/shut down processes; Blast furnace operation and its irregularities: deadman's zone, hanging, scaffold, pillering; Limitations of Blast furnace Iron production; Sponge Iron Productions: Using gases as reducing agent: Midrex process, HYL, Using solid as reducing agent process: SL/RN process; Smelt Iron Reduction Methods: COREX, INRED, ELRED; Burden/charge/mass balance calculations.

Unit -4: LD converter steel making

Pre-treatment; Role of slag; steelmaking reactions; Linz-Donawitz (LD) convertor steelmaking: LD design, raw material, chemical reactions in LD: Thermodyanamics of O, S, P, C removal; Modren developments in LD convertor.

Unit -5: EAF steel making

Electric arc furnace steelmaking (EAF-SM): EAF Design, raw materials, EAF steel making process; Modern developments in EAF-SM.

Continuous casting process (CCP): Cast into semi-finished products; Grain structure; Heattransfer zones in CCP; Modern developments in CCP;

Burden/charge/mass balance calculations in iron and steel making.

Course outcomes: The student will be able to

- Get exposure to the history of iron and steel making methods
- Understand the physical chemistry of iron making
- Understand the sponge iron making process
- Understand the physical chemistry of steel making
- Understand the operations of an integrated steel plant
- Understood the process of continuous casting of steel
- Balance materials in iron and steel making

Text books:

- Ahindra Ghosh & Amit Chaterjee, Iron making and Steel making, 2008
- Bashforth G.R, Manufacture of Iron and Steel, Volumes I - IV, 1996

References:

- Dr. R.H. Tupkary and V.H. Tupkary, Modern Steel making, 1998
- Dr. R.H. Tupkary and V.H. Tupkary, Modern Iron making, 1998
- K. Chakrabarthi , Steel Making, 2007
- Turkdogen, Steel making, 2000
- Bodsworth, Physical Chemistry of Iron & Steel, 1999
- Dutta S.K., Lele A.B., Metallurgical Thermodynamics Kinetics and Numericals

Category: Program Core Course Lab Subject code:MM2801

Mechanical Properties of Materials Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0-0-3-1.5

Course Objectives:

- To provide hands on experience on various mechanical testing procedures.
- To study various deformation mechanisms.
- Investigate mechanical properties of ferrous and nonferrous metals.
- To analyze results and draw conclusions from the tests.

List of experiments:

1) Brinell Hardness testing of ferrous and non-ferrous samples.
2) Rockwell hardness testing of ferrous and non-ferrous samples.
3) Vickers Hardness testing of ferrous and non-ferrous samples.
4) Analysis of tensile testing data.
5) Calculate True stress Vs True stain and compare with engineering stress strain curve.
6) Compression test of a brittle material.
7) Three point bend test of a Rebar.
8) Erichsen cupping test for ductility measurement of non-ferrous metals.
9) Charpy and Izod test (V & U Groove notch) at room temperature.
10) Establishment of the ductile - brittle transition temperature of the material.
11) Analysis of Creep testing data
12) Analysis of Fatigue testing data

Course outcomes: The student will be able to

- Operate testing equipment for measuring mechanical properties
- Extract, interpret and analyse data from mechanical testing.
- Design and select metals for engineering applications.
- Derive relationship between mechanical properties of metals.

Textbooks:

- Mechanical Behavior of Materials Laboratory Manual.
- Mechanical Testing and Evaluation, ASM Metals Handbook, Vol 08.
- Hardness Testing Principles and Applications, ASM International, Konrad Herrman, 2011.
- Dieter, G, E., “Mechanical metallurgy (SI metric edition)”, McGraw-Hill, 1988.

Category: Program Core Course Lab Subject code:MM2802

Metal Casting and Joining Lab

Externals: 60 Marks L-T-P-C*

Internals: 40 Marks 0-0-3-1.5

Course Objectives:

- To provide hands on experience on casting and welding operations
- To understand and analyze different Casting and welding defects.
- To determine green sand mould properties
- Casting of metals and alloys

List of experiments:

1) preparation of Green sand mold
2) To determine the permeability number of the sand mould specimen
3) To determine the shear strength of the sand mold specimen
4) To determine the moisture content in the sand mold specimen
5) To determine the hardness of the sand mold specimen
6) To demonstrate melting of Aluminum in Induction furnace
7) To cast aluminum using green sand mould
8) To weld the mild steel samples by manual metal arc welding process and visually inspect thedefects along with the microstructure variations.
9) To weld the mild steel samples by metal inert gas welding process and visually inspect the defectsalong with the microstructure variations.
10) To weld the mild steel samples by oxy acetylene gas welding process and visually inspect thedefects along with the microstructure variations.
11) To weld the aluminum samples by tungsten inert gas welding process and visually inspect thedefects along with the microstructure variations.
12) To weld two similar metals by friction stir welding process and visually inspect the defects alongwith the microstructure variations

Course outcomes: The student will be able to

- Describe various casting and welding operations
- Determine green sand mould properties
- Demonstrate melting and casting of metals/alloys
- Analyze different Casting defects.
- Analyze different welding defects

Textbooks:

- R. W. Heine, C. R. Loper, P. C. Rosenthal, ”Principles of metal casting”, Mc Graw Higher Ed, 1976.
- P. K. Jain, “Principles of foundry technology”, Mc Graw-Hill 1987.
- K. Easterling, “Introduction to physical metallurgy of welding”, Butterworth-Hienemann 1992.
- Sindo Kou “Welding Metallurgy”, John Wiley & sons Publications 2003

Category: Humanities and Social Sciences course Subject code: HS3101

Corporate Communication

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0-0-2-1

Course Objectives:

- To make the students efficient communicators via experiential learning.
- To enhance learners' analytical and creative skills, so that they will be capable to address a widevariety of challenges in their professional lives.
- To help learners to improve the leadership qualities and professional etiquette
- To expose learners to an effective communicative environments.

Unit-1: Introduction to communication

Introduction, Importance of Communication Skills, Definition, Scope and Nature, Verbal and Nonverbal communication

Unit-2: Reading Skills

Reading Comprehension of unseen passage, Prose, News Paper Reading and Analysis (Editorial), Novels, different research articles, crack the answers in the unseen paragraphs in the competitive exams etc….

Unit-3: Functional Grammar

1. Parts of Speech (Functional Usage)
2. Subject and predicate (useful at workstations)
3. Conjunctions-Gap fillers (Linkers; connectors; cohesive devices)
4. Verbs & Verb patterns (Transitive and Intransitive - Finite and Infinite - Regular and Irregular – Modals)
5. Tenses (Various Applications)
6. Prepositions/ Prepositional verbs (idiomatic expressions, one word substitutions, phrasal verbs)
7. Adjectives (describing, narrating, effective presentation)

Unit-4: Enhancing Vocabulary

Developing Professional vocabulary, different forms of verbs (verb forms: v1, v2, v3) – Using Dictionary: Spelling, jargon specific vocabulary, context specific vocabulary, right choice of vocabulary etc.

Unit-5: Composition

Paragraph, Essay, Expansion, Describing the Pictures, Giving Directions, Situational Dialogue writing, Social and Professional Etiquette, Telephone Etiquette, email etiquettes, Role plays, JAM, and elocution etc.

Course outcomes: The student will be able to

- develop interpersonal communication, small group interactions and public speaking.
- develop confidence and skills related reading comprehension.

- improve a logical framework for the critical analysis of spoken, written, visual and mediatedmessages upon diverse platforms.

- demonstrate the ability to apply vocabulary in practical situations.

Textbooks:

1. Joseph Mylal Biswas book of English Grammar
2. R. Murphy -Cambridge Press
3. Wren and Martin
4. The Good Grammar book by OUP
5. Communication skills by M. Raman and Sangeeta Sharma
6. How to Win Friends and Influence people by Dale Carnigie
7. How to Read and Write Better by Norman Lewis
8. Better English by Norman Lewis
9. Use of English Collocations by OUP
10. www.humptiesgrammar.com
11. www.bbcenglisgh.com
12. www.gingersoftware.com
13. www.pintest.com

Category: Program Core Course Subject code: MM3101

Heat Treatment

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- To study and understand heat treatment process of steels
- To understand basic principles of phase transformation during heat treatment of steels.
- To study and understand heat treatment process of alloy steels and cast Irons
- To study and understand different surface and thermo mechanical treatments
- To study and understand heat treatment process of non-ferrous metals and alloys

Unit -1: Introduction

Heat treatment processes for steels: stress relieving, annealing, spherodizing, normalizing, hardening, tempering and hardenability, austempering, martempering, sub zero treatment, patenting.

Unit -2: Principles of heat treatment of Steels

formation of austenite on heating, decomposition of austenite, ttt curves, pearlitic transformation, bainitic transformation and martensitic transformation.

Unit -3: Heat treatment of commercial Steels and Cast Irons

Alloy steels, Structural and Tools steels, Grey Cast Iron, White Cast Iron, Malleable Cast Iron, SG Cast Iron, and Alloy Cast Irons.

Unit -4: Surface and Thermomechanical treatments

Surface heat treatment, Carburizing, Cyaniding, Nitriding Flame and Induction hardening, Thermo mechanical treatments: HTMT, LTMT, ausforming, isoforming, cryoforming.

Unit -5: Heat treatment of Non Ferrous Metals and Alloys

Heat treatment of non-ferrous alloys: Copper and its alloys, Aluminum and its alloys, Titanium, nickel and its alloys.

Course Outcomes: The student will be able to

- Explain the heat treatment process of steels
- Determine the principles of phase transformation during heat treatment of steels.
- Explain the heat treatment process of alloy steels and cast Irons
- Explain different surface and thermo mechanical treatments
- Explain the heat treatment process of non-ferrous metals and alloys

Textbooks:

- Rajan, T. V and Sharma, C. P., "Heat treatment principles and techniques (2nd edition)", Pren- tice hall of India, 1994.
- Reed-Hill, R, E and Abbaschian, R., "Physical Metallurgy Principles (3rd edition)", PWS-Kent publishing company, 1994.

Reference:

- William. E. Bryson, “Heat treatment: Master control manual”, Carl Hanser Verlag, Munich, 2015.
- ASM Handbook “Volumes 4A, 4B and 4E of Heat Treatment”, ASM International, 2016.

Category: Program Core Course Subject code: MM3102

Non Ferrous Extractive Metallurgy

Externals: 60 Marks L-T-P-C*
Internals: 40Marks 3-1-0-4

Course Objectives:

- To understand the methods of metal extraction and sources of non ferrous metals.
- To study the thermodynamics principles involved in metal extraction.
- To study the methods of extracting non ferrous metals from their ores.
- To study various methods of refining the extracted metals.

Unit-1: Introduction

History of non ferrous Metals: Early developments in metal extraction (introduction, discovery of metals and their importance, important land marks, non ferrous metals in Indian history, uses of non ferrous metals), sources of non-ferrous metals (sources in land and sea, exploration methods, methods of beneficiation, non ferrous metals wealth in India).

Unit-2: Basic principles of extraction

Principles of metals extraction, (thermodynamic principles, homogeneous and heterogeneous reactions, Ellingham diagrams, kinetic principles, principles of electro-chemistry), general methods of extraction,(pyro-metallurgy calcinations, roasting and smelting, hydrometallurgy - leaching, solvent extraction, ion exchange, precipitation, and electrometallurgy - electrolysis and electro - refining), general methods of refining, (basic approaches, preparation of pure compounds, purification of crude metal produced in bulk).

Unit-3: Extraction of metals from oxides

Extraction of metals from oxide sources: Extraction of metals such as magnesium, aluminum, tin.

Unit-4: Extraction of metals from sulphides

Extraction of metals from sulphide ores: Pyro - metallurgy and hydro - metallurgy of sulphides, production of metals such as copper, lead, zinc, gold, silver)

Unit-5: Extraction of metals from halides

Production of halides and refining methods, production of reactive and reactor metals. Methods of extraction of metals such as titanium, rare earths, uranium, thorium, plutonium, beryllium, zirconium

Course outcomes: Students will be able to

Explain the significance of Ellingham diagram for extraction of metals.

Identify the suitable techniques to extract metals form ore.

Explain the various extractive metallurgy routes.

Apply principles of thermodynamics for extraction of metals.

Textbooks:

- H S Ray, Extraction of Non-Ferrous Metals, K P Abraham and R. Sridhar.

- G B Gill, Non Ferrous Extractive Metallurgy”, John Wiley & Sons.

Category: Program Core Course Subject code: M3103

Mechanical Working of Metals

Externals: 60 Marks L-T-P-C*

Internals: 40 Marks 3-1-0-4

Course Objectives:

- To understand the basic concepts of stress and strain in different materials
- To study about the cold, warm and hot working processes
- To study and analyze the drawing of metals working such as extrusion and drawing
- To study and analyze various sheet metal forming processes like bending and blanking
- To understand the other mechanical working processes

Unit-1: Plastic and Elastic behaviour

Stress-strain relations in elastic and plastic deformation, concept of flow stress, deformation mechanisms, basic metal working concepts and plasticity, yield criterion, role of temperature and friction in metal working.

Unit-2: Cold and hot working processes

Hot, warm and cold working, forging, rolling-types, analysis, parameters affecting the process, defects, theircauses and remedial measures.

Unit-3: Metal drawing process

Extrusion, wire and tube drawing-types, analysis, parameters affecting the process, defects, their causes andremedial measures.

Unit-4: Metal forming process

Sheet metal working processes; classification of metal forming processes, estimation of force and energy requirements; metals used in press working, lubrication; theory of shearing, blanking, piercing, bending and stretch forming

Unit-5: Other metal forming processes

Deep drawing, hydro forming, coining and embossing - types, parameters affecting the process, lubrication in metal forming processes, defects in various sheet metal forming processes, their causes and remedial measures

Course outcomes:

A student shall be able to,

- Apply the concepts of stress-strain in the design of various deformation processes.
- Classify the metal working processes
- Can explain principle of forging, determination of forging load & its application
- Understand the manufacturing of tubes and pipe by extrusion method
- Acquainted with sheet metal working
- Gain knowledge on various conventional and advanced deformation operations.
- Identify appropriate deformation operation for a particular material

Textbooks:

- George E. Dieter, Mechanical Metallurgy, 2017
- Amit Bhaduri , Mechanical Properties and Working of Metals and Alloys, 2019

References:

- J.P. Kaushish, Manufacturing processes, 2000
- Swadesh Kumar Singh, Production Engineerin, 1997

Category: Program Elective Course Subject code:MM3104

Transport Phenomena

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- To introduce the concepts of fluid flow, heat transfer and mass transfer.
- To understand transport properties of fluids and solids.
- To understand the fundamental phenomena, governing equations and assumptions used in the analysis of transport processes.
- To predict the boundary conditions for various systems.

Unit-1: Introduction

Balance of quantities using elemental volume approach, continuity equation.

Unit-2: Kinetics

Newton's law of viscosity, Navier - Stokes equation, laminar flow problems, exact solutions in rectan- gular, cylindrical and spherical coordinate systems. Friction factors, correlations for turbulent regime, Darcy's law, flow through porous media.

Unit-3: Heat transfer

Fundamentals of heat conduction, convection, radiation and their combined effect; steady and unsteady heat transfer, exact analytical solutions, correlations for conjugate heat transfer.

Unit-4: Mass transfer

Diffusion and its application in solid state, convective mass transfer, unsteady diffusion in finite and infinite bodies, diffusion and chemical reactions.

Unit-5: Other phenomena

Coupled phenomena in transport, non-dimensional numbers and their correlations of different regimesand analogies.

Course outcomes: The student should be able to

- Apply mass and momentum transfer principles in the extraction and refining of metals.
- Apply the heat transfer principles in furnace design and thermo-mechanical treatment of materials.
- Calculate material and energy balances in material flow systems.
- Illustrate simultaneous momentum, heat and mass transfer in metallurgical reactions.
- solve simple partial differential equations relevant to transport phenomena

Textbooks:

- G. H. Geiger, D. R. Poirier, Transport Phenomena in Materials Processing, John Wiley& Sons, 2010 .
- K. Mohanty, Rate Processes in Metallurgy, PHI, 2012

- James R. Welty, Charles E. Wicks, Robert E. Wilson, Gregory L. Rorre, “Fundamentals of Momentum, Heat and Mass Transfer”, 5th Edition, John Wiley & Sons, 2007.

References:

- David R Gaskell, An Introduction to Transport Phenomena in Materials Engineering, Momentum Press, 2013.
- R. Byron Bird Warren E. Stewart Edwin N. Lightfoot, “Transport Phenomena”, John Wiley & Sons, 2002.
- G. S. Upadhyaya and R. K Dube, Problems in Metallurgical Thermodynamics and Kinetics, Pergamon, NewYork, 1982
- D. R. Poirier , G. H. Geiger,”Transport Phenomena in Materials Processing”, The Minerals, Metals & Materials Series, Springer Cham, 2016.

Category: Program Core Course Lab Subject code:MM3701

Heat Treatment Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0-0-3-1.5

Course Objectives:

- To introduce various kinds of heat treatment operations to the students
- To study the effect of heat treatment on the microstructure of the metals
- To study the effect of heat treatment on the mechanical properties of the metals
- To study and understand the age hardening of aluminum alloy
- To study the effect of hardenability of mechanical property of steel

List of experiments:

1) Annealing of medium carbon steels (AISI C105, EN8, EN24) and observation of microstructure.
2) Normalizing of medium carbon steels (AISI C105, EN8, EN24) and observation of microstructure.
3) Hardening of medium carbon steels (AISI C105, EN8, EN24) and observation of microstructure.
4) Study of tempering characteristics of water quenched steel.
5) Study of age hardening phenomena in duralumin.
6) Spheroidizing of a given high carbon steel.
7) Determination of hardenability of medium carbon steel by Jominy end quench Test.
8) Effect of hardening heat treatment on the hardness of cast iron (Grey and Ductile Cast iron)

Course Outcomes: The student should be able to

- perform various heat treatment operations on steels
- co-relate microstructure and mechanical properties
- determine the effect of heat treatment on the mechanical properties of the metals
- determine the age hardening effect on aluminum alloy
- determine the effect of hardenability of mechanical property of steel

Textbooks:

- Rajan, T. V and Sharma, C. P., “Heat treatment principles and techniques (2nd edition)”, Pren- tice hall of India, 1994.
- Reed-Hill, R, E and Abbaschian, R., “Physical Metallurgy Principles (3rd edition)”, PWS-Kent publishing company, 1994.

Category: Program Core Course Lab Subject code: MM3702

Extractive Metallurgy Lab

Externals: 60 Marks L-T-P-C*
Internals: 40Marks 0-0-3-1.5

Course Objectives:

- To understand the extraction methods of ferrous and non ferrous metals.
- To understand the physical and chemical beneficiation of metals from their ores.
- To compare different extraction processes and deduce the most effective one
- To correlate laboratory and industrial extractions methods

List of Experiments:

1) Calcination of lime and study its kinetics
2) Roasting Copper sulphate
3) Preparation of iron by reaction of iron(III) oxide with aluminum (thermite reaction)
4) Electrorefining of copper
5) Electrowinning of copper
6) Electrowinning of Nickel
7) Leaching of chalcopyrite by NaOH+Na2CO3 mixture
8) Iron production from pyrometallurgical route
9) Solvent extraction process
10) Ion exchange process
11) Zone refining process

Course outcomes: The student should be able to

- gain the knowledge and experience on some of the mineral beneficiation and extraction operations of minerals.
- study roasting, calcination and smelting of ores
- electro plating and electro-chemical refining of metal
- visualize industrial operations which are a part of the extraction of ferrous and non-ferrous metals

Textbooks:

- H S Ray, **Extraction of Non-Ferrous Metals**, K P Abraham and R. Sridhar.
- G B Gill, Non Ferrous Extractive Metallurgy", John Wiley & Sons.

Category: Humanities and Social Sciences course Subject code: HS3201

Soft Skills

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0-0-2-1

Course Objectives:

- To enable students speak effectively in formal and informal situations
- To equip the students with necessary writing skills in order to face the corporate world
- To strengthen the writing skills of the students and help them in documentation
- To enable students sharpen their communication skills towards writing a persuasive resume andeffective job application letters
- To equip students with pre-presentation steps, to understand the structure of a good presentation, and devise various techniques for delivering a successful presentation
- To make students understand the importance of team work and group presentations and groupdiscussions

Unit-1: Activities on Fundamentals of Inter-personal Communication

Starting a conversation - responding appropriately and relevantly, using the right body language, Role Play in different situations & Discourse Skills using visuals.

Unit-2: Activities on Reading Comprehension

General Vs Local comprehension, reading for facts, guessing meanings from context, scanning, skimming, inferring meaning-critical reading, surfing Internet

Unit-3: Activities on Writing Skills

Structure and presentation of different types of writing- Resume writing/ e-correspondence/ Technical report writing- planning for writing, improving one's writing.

Unit-4: Activities on Presentation Skills

Oral presentations (individual and group) through JAM sessions/seminars/PPTs and written presentations

Unit-5: Activities on Group Discussion, Debate and Interview Skills

Dynamics of group discussion- intervention- summarizing-modulation of voice-body language-relevance-fluency and organization of ideas and rubrics for evaluation- Concept and process-pre-interview planning-opening strategies-answering strategies- interview through tele-conference & video-conferencing - Mock Interviews.

Course outcomes: The student should be able to

- communicate effectively in formal and informal situations

- understand the structure and mechanics of writing resumes, reports, documents and e-mails
- present effectively in academic and professional contexts
- develop communication in writing for a variety of purposes
- identify areas of evaluation in Group Discussions conducted by organizations as part of theselection procedure
- overcome stage fear and tackle questions

Textbooks:

- Soft Skills Training: A workbook to Develop Skills for Employment by Frederick H. Wentz
- Everyone Communicates, Few People Connect: What the Most Effective People do Differently by John C. Maxwell
- How to Talk to Anyone: 92 Little Tricks to Have Big success in Relationships by Leil Lowndes
- Teamwork101: What Every Leader Needs to Know by John C. Maxwell
- AdaptAbility: How to Survive Change You Didn't Ask For by M.J. Ryan
- Conflict Communication: A New Paradigm in Conscious Communication by Rory Miller

Category: Program Core Course | Subject code: MM3201

Materials Characterization

Externals: 60 Marks | L-T-P-C*
Internals: 40 Marks | 3-1-0-4

Course Objectives: To understand

- The basic concept of different characterization techniques and optical microscope
- To analyze and understand the behavior of materials from characterization techniques.
- To study crystal structure, chemical composition, phase, residual stress and texture of materials.
- To study the microstructure of materials from optical and electron microscopes.
- To understand the spectroscopic, thermal and electrical characterization techniques.

Unit-1: Optical Microscope

Introduction, scope of subject, classification of techniques for characterization, macro & micro-characterization structure of solids, Metallographic techniques: Optical metallography, image analysis, quantitative phase estimation.

Unit-2: Electron Microscope

Electron optical methods: Scanning electron microscopy and image formation in the SEM, Transmission electron microscopy (TEM), Scanning tunneling microscopy (STM), Atomic force microscopy (AFM)and scanning transmission electron microscopy (STEM).

Unit-3: XRD

Diffraction methods: X - ray diffraction, crystal systems and space groups, Bravais lattices, direct and reciprocal lattice, Braggs law, powder diffraction and phase identification, single crystal diffraction, structure factor, X-ray crystal structure determination, Residual stress analysis.

Unit-4: Spectroscopy

Optical & X -ray spectroscopy: atomic absorption spectroscopy, X -ray spectrometry, infrared spectroscopy, Raman spectroscopy, EDS and WDS.

Unit-5: Thermal Analysis

Bulk averaging techniques: Thermal analysis, DTA, DSC, TGA, TMA, dilatometry, resistivity/ conductivity.

Course outcomes: The student will be able to

- Familiar on different characterization techniques and optical microscope.
- Describe the principles of optical and electron microscopy.
- Explain the principles of XRD for materials characterization.
- Understand Spectroscopic techniques for characterize materials.
- Analyse thermal reactions using thermal analysis data.

Textbooks:

- Spencer Michael, Fundamentals of Light Microscopy, Cambridge University Press, 1982

- David B. Williams and C. Barry Carter, Transmission Electron Microscopy: A Textbook for Mate rials Science, Springer, 2009.
- Joseph I Goldstein, Dale E Newbury, Patrick Echlin and David C Joy, Scanning Electron Microscopy and X-Ray Microanalysis, Springer, 2005.

References:

- B.D.Cullity and S.R.Stock, Elements of X-Ray Diffraction, Prentice Hall, 2001.
- G.W.H. Hohne, W.F. Hemminger, H.J. Flammersheim, Differential Scanning Calorimetry, Springer, 2003.
- Douglas B. Murphy, Fundamentals of light microscopy and electronic imaging, Wiley, 2001.
- David G. Rickerby, Giovanni Valdrè, Ugo Valdrè, Impact of Electron and Scanning Probe Microscopy on Materials Research, Springer, 1999.

Category: Program Core Course Subject code: MM3202

Corrosion Engineering

Externals: 60 Marks L-T-P-C*

Internals: 40Marks 3-1-0-4

Course Objectives:

- To understand the technological importance of corrosion studies
- To study types and basic concepts of corrosion
- To study and understand the kinetics of corrosion
- To study and understand the preventive measures of corrosion

Unit-1: Introduction

Technological importance of corrosion study - introduction to corrosion, definition, learning objectives, degradation process-mechanical and chemical process. Dry corrosion and wet corrosion. Local and uniform corrosion. Cost of corrosion-direct loss and indirect loss, cost of corrosion in various industries.

Unit-2: Basic Concepts of Corrosion

Electochemical principles of corrosion - cell analogy, cathode, anode, electrolyte, cathodic and anodic reactions, types of corrosion cell. Concept of free energy (driving force of corrosion based on thermodynamical studies), cell potential and emf, Nernst equation and their application on corrosion. Concept of single electrode potential, reference electrodes, half cell reaction, types of reference electrode-SHE, Ag-AgCl, SCE, Cu-$CuSO_4$.Emf and galvanic series-their uses in corrosion studies. Eh-pH diagrams-fundamental aspects. Construction of Eh-pH diagrams.

Unit-3: CorrosionKinetics

Corrosion rate expressions-Faraday's law, area effect, weight loss, thickness loss. Electrode – solution interface – over potential, definition and types of polarization-factors affecting them. Exchange current density-polarization relationships. Mixed potentials-concepts and basics. Mixed potential theory-mixed electrodes (bimetallic couples), activation and diffusion controlled processes. Application of mixed potential theory. Corrosion rate measurements (determination).Passivity-definitions and influencing parameters. Passivity-design of corrosion resistant alloys, factors affecting passivity.

Unit-4: TypesOfCorrosion

Different forms of corrosion Mechanism, characteristic features, causes and remedial measures of uniform corrosion, galvanic corrosion, crevice corrosion. Pitting corrosion, intergranular corrosion(including weld decay & knife-line attack). Erosion corrosion, selective leaching and stress corrosion cracking. Hydrogen damage-types, characteristics, mechanism and preventive measures.

Unit-5: Cathodic Protection and Coating

Principles of corrosion prevention-material selection, control of environment including inhibitors. Cathodic protection-principle, classification, influencing factors and design aspects. Anodic protection-principle, influencing factors and design aspects. Coatings and design considerations (corrosion prevention).

Course outcomes: The student should be able to

- Understand electro chemical fundamentals
- Understand corrosion preventing methods
- Understand environmental induced corrosion
- Solve corrosion problems

Textbooks:

- Fontana. “Corrosion Engineering”,
- Zaki Ahmad “Principles of Corrosion engineering and corrosion control”,
- Pierre R. Roberge. “Handbook of Corrosion Engineering”,

Category: Program Core Course Subject code: MM3203

Powder Metallurgy and Additive Manufacturing

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-0-0-3

Course Objectives:

- To understand the basics and overview of powder metallurgy
- To study various powder production methods and analyzing the characteristics of powders
- To understand various cold and hot compaction methods
- To study various sintering and post sintering processes.
- To study various additive manufacturing processes.

Unit-1: Introduction

Introduction to powder metallurgy, steps in powder metallurgy, advantages and limitations of powder metallurgy, recent trends.

Unit-2: Fabrication and Characteristics

Powder production methods: mechanical, chemical, atomisation and physical methods, powder treatment and handling.

Particle size & shape distribution, electron microscopy of powder, interparticle friction, packing and flow characteristics of powders, density compression ability, powder structure, chemical characterization

Unit-3: Powder Shaping

Particle packing modifications, lubricants and binders, powder compaction and process variables, pressure and density distribution during compaction, isostatic pressing, injection molding, powder extrusion, slip casting, tape casting, analysis of defects of powder compact.

Unit-4: Sintering

Theory of sintering, sintering of single and mixed phase powder, liquid phase sintering, sintering variables, modern sintering techniques, physical & mechanical properties evaluation, structure-property correlation study, defects analysis of sintered components, post sintering operations.

Application of Powder Metallurgy: Filters, Tungsten Filaments, Self-Lubricating Bearings, Porous Materials, ODS Alloys.

Unit-5: Additive Manufacturing

Introduction to AM, advantages of AM, steps in AM, classification of AM processes and types of materials for AM.

Course outcomes: The student should be able to

- Understand the need for pm parts.
- Understand the powder production and characterization techniques.
- Understand various powder shaping techniques.
- Understand the role of binders and lubricants in pm.

- Understand the significance of various stages in sintering and influence of sintering atmospheres.
- Uptimize process parameters of the powder making operations.
- Have a basic knowledge of additive manufacturing.

Textbooks:

- R.M. German, "Powder Metallurgy Science", Metal Powder Industry, 1994.
- Anish Upadhyaya, G.S. Upadhyaya, "Powder Metallurgy science, technology and materials", Universities Press, 2011.
- Damir Godec, Joamin Gonzalez-Gutierrez, Axel Nordin, Eujin Pei, Julia Ureña Alcázar, "A Guide to Additive Manufacturing", Springer Tracts in Additive Manufacturing, 2022.

References:

- P.K. Samal and J. W. Newkird, "Powder Metallurgy", ASM Handbook, Volume 7,2015.
- R. M. German, "Powder Metallurgy & Particulate Materials Processing", MPIF, 2005.
- J. S. Hirschhorn, "Introduction to Powder Metallurgy", American Powder Metallurgy Institute, 1976.
- P. C. Angelo and R. Subramanian: Powder Metallurgy- Science, Technology and Applications, 2008.

Category: Program Core Course Lab Subject code:MM3801

Materials Characterization Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0-0-3-1.5

Course Objectives:

- To familiarize in microstructure feature
- To understand mode of fracture
- To provide hands on experience of XRD, FESEM, and EDX
- To study and analyze the peaks of XRD of materials
- To analyze quantitatively the chemical composition of material
- To analyze the microscopic images of materials produced by FESEM

List of experiments:

1) Image analysis of microstructures using OM.
2) Microstructural analysis using SEM.
3) Fractography analysis using SEM.
4) Chemical analysis of phases using SEM-EDS
5) Index and calculate the lattice parameter of cubic systems.
6) Index and calculate the lattice parameter of Noncubic systems.
7) Calculate the precise lattice parameter.
8) Calculate the crystalline size, lattice strain and residual stress from XRD data.
9) Phase identification from given XRD data.
10) Oxidation kinetics using TGA / DTA data.
11) Determination of thermal properties using DSC data.
12) Microstructural analysis using TEM image.

Course outcomes: The student will be able to

- Quantify microstructural features using image analysis tools.
- Identify the modes of fracture.
- Analyse the chemical composition of materials through sem-eds.
- Determine crystal structure of the material using xrd technique.
- Determine the thermal properties of materials.

Textbooks:

- S Zhang, L. Li and Ashok Kumar, Materials Characterization Techniques, CRC Press, 2008.
- David B. Williams, C.Barry Carter, Transmission Electron Microscopy, Textbook for Materials Science, Springer, 2009.
- Khangaonkar P R, Penram, An Introduction to Material Characterization, Intl. Publishing (India) Pvt. Ltd, mumbai, 2010.

References:

- B.D.Cullity and S.R.Stock, Elements of X-Ray Diffraction, Prentice Hall, NJ, 2001.
- ASM Handbook, Vol.10, Materials Characterization, ASM International, USA, 1998.

Category: Program Core Course Lab Subject code: MM3802

Corrosion Engineering Lab

Externals: 60Marks L-T-P-C*
Internals: 40Marks 0-0-3-1.5

Course Objectives:

- To measure the corrosion rate of two different metals and to establishing corrosion mechanisms.
- Defining corrosion resistance of materials and how to develop new corrosion resistant alloys and to Estimating service life of equipment.
- Developing corrosion protection processes.
- To show the effectiveness of the use of inhibitors.
- Defining the critical potential values for materials in various environments.

The list of experiments:

1) Weight loss-corrosion rate measurement
2) Effect of inhibitor on rate of corrosion (inorganic inhibit or ororganic inhibitor)
3) Crevice & Pitting corrosion testing
4) Corrosion protective coatings (Hot-dip galvanization)
5) Corrosion protective coatings (Anodization)
6) Corrosion protective coatings (Electroplating or electroless plating)
7) Corrosion prevention Protective coatings (hardness test by pencil test)
8) Corrosion prevention Protective coatings (immersion test)
9) Corrosion prevention Protective coatings (salt spray test)
10) Corrosion rate measurement using Tafel extrapolation or potentio-state
11) Construction and interpretation of E-Ph diagram

References:

- ASTMG1—Standard practice for preparing tests specimens.
- ASTM G31-72—Standard practice for laboratory immersion corrosion testing of metals.
- ASTMG48-11—Standard test methods for pitting and crevice corrosion resistance of stainless steels and related alloys by use of ferric chloride solution.
- ASTMD336305- Standard test method for film hardness by pencil test.
- ASTMD 6943-03— Standard practice for immersion testing of industrial protective coatings.
- ASTM G 36 94—Standard practice for evaluating stress-corrosion-cracking resistance of metals and alloys in a boiling magnesium chloride solution
- ASTMG3-14—Standard practice for conventions applicable to electrochemical measurements in corrosion testing.\
- ASTMA262-15—Standard practices for detecting susceptibility to intergranular attack in austenitic stainless steels
- ASTMG514—Standard reference tests method for making Potentiostat anodic polarization measurements.
- ASTMG59-97—Standard test method for conducting Potentiostat polarization resistance measurements.

Category: Humanities and Social Sciences course Subject code: BM4101

Fundamentals of Management for Engineers

Externals: 60 Marks	L-T-P-C*
Internals: 40 Marks	3-0-0-3

Course Objectives:

- Identify and apply appropriate management techniques for managing business.
- Have a conceptual knowledge about the planning and decision making.
- Apply the concept of organizing for the effective functioning of a management.
- Evaluate the concepts of staffing & directing.
- Demonstrate the techniques for controlling and coordination.

Unit-1: Nature of management

Nature of management, importance, functions of management, role of manager, evolution of management thoughts.

Unit-2: Planning

Nature, importance, types, steps, limitations, decision making, types, process of rational decision making, techniques of decision making.

Unit-3: Organizing

Concept, nature, process, purpose and significance, authority and responsibility, delegation of authority, centralization and decentralization, departmentalisation.

Unit-4: Staffing and directing

Meaning, importance of recruitment and selection, training and development;
motivation: meaning, nature; leadership: meaning and styles; communication: nature, process and barriers.

Unit-5: Controlling

Need, importance and process, effective control systems, techniques, traditional and modern coordination, need and importance.

Course outcomes: The student should be able to

- identify and apply appropriate management techniques for managing business.
- have a conceptual knowledge about the planning and decision making.
- apply the concept of organizing for the effective functioning of a management.
- evaluate the concepts of staffing & directing.
- demonstrate the techniques for controlling and coordination.

References:

- Principles of Management", Pagare Dinkar.
- Principles and Practice of Management", L. M. Prasad.
- Principles and Practice of Management", Satya Narayan and Raw VSP.
- Management Principles and Practice", Srivastava and Chunawalla.

Category: Program Core Course Subject code:MM4101

Computational Materials Engineering Lab

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 0-0-2-1

Course Objectives: To make the students familiarize with

- various kinds of models used in computational materials engineering
- application of various numerical techniques to solve the equations in the models
- coding and simulation tools
- script coding and latex programming

The list of experiments:

1) Introduction to the programming using C or Python
2) Various visualization tools for the numerical data
3) Analyzing the cohesive strength using script code
4) Ring compression test using finite elemental method
5) Basics of MATLAB programming
6) Programs using MATLAB
7) Report writing using Latex coding
8) Basics of Molecular Dynamics simulation
9) Calculate the toughness of the material using the tensile test data
10) Simple engineering drawing using CAD tools
11) Analysis of microstructure using software tools
12) Analysis of data using linear regression for one/multi dimensional data
13) Solving the given set of linear and nonlinear equations
14) Find the derivative of the given numerical data using appropriate numerical methods
15) Find the integral of the given data using appropriate numerical methods
16) Solving the Laplace equation using finite divided differences
17) Solving the diffusion equation using finite divided differences
18) Find the equilibrium concentration of the components in a reactor for an open system
19) Find equilibrium concentration of the phases modeled using regular solution model at the given temperature for binary system
20) Calculate the thermodynamic properties of the given phase as a function (i) temperature and (ii) composition
21) Calculate steady state temperature profile of a plate whose edges are maintained at constant temperature
22) Calculate the variation of temperature profile of a rod whose end is maintained in a pit furnace

Course outcomes: The student should be able to

- develop a simple model for metallurgical problems
- use appropriate techniques to solve the model developed
- implement simple codes/ programs in one of the programming languages
- analyze the data generated

Suggested Textbooks and References:

- Chapra, S. C., & Canale, R. P. (2016). "Numerical Methods for Engineers", McGraw-Hill Edu- cation.
- Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. (2007). "Numerical Recipes: The Art of Scientific Computing (3rd ed.)", Cambridge University Press.
- Lee, J. G. (2011). "Computational Materials Science: An Introduction", CRC Press.
- Ohno, K., Esfarjani, K., & Kawazoe, Y. (2018). "Computational Materials Science", Springer Berlin Heidelberg.

Category: Program Elective Course | Subject code: MM3221

Secondary steel making

Externals: 60 Marks | L-T-P-C*
Internals: 40 Marks | 3-0-0-3

Course Objectives:

- To study and understand various refining techniques of steel making
- To study and understand various vacuum degassing methods
- To understand the metallurgy involved in tundish used for storing refined steel before continuous casting
- To understand the Laddle, Tundish and Injection metallurgy process
- To study and understand various secondary treatment methods of steels
- To learn the numerical problems of steel making

Unit-1: Classification of secondary steel making

Ladle metallurgy, Injection metallurgy and Tundish metallurgy; Activity and composition relationships, Thermodynamics of steel making: Raoults and Henry's law; Deoxidation, Desiliconisation, Demanganesisaton, Decarburisation, Desulphurisation, Dephosphirisation;

Unit-2: Vacuum degassing methods

VOD (vacuum oxygen decarburizer), AOD (Argon oxygen decarburisation), RH and DH degassing of O, H, N from liquid steel; EAF+AOD (Electric Arc Furnace+Argon oxygen decarburisation) duplexing process, EAF+AOD+VOD triplexing process of stainless steel

Unit-3: Clean steel technology

Inclusion engineering, Tundish design, Inclusions modifications, Remelting process: VAR (Vacuum arc remelting), ESR (Electro-slag Remelting)

Unit-4: Secondary treatment of Alloy steels

Production scheme of Dual Phase steels, TRIP Steels, TWIP steels, Interstitial-Free Steels,Bake hardened steels, HSLA Steels

Unit-5: Numerical problems of steel making

Material/charge balance in steel making, Thermodynamics of steel making, deoxidation problems, Raoults and Henry's law based problems

Course Outcomes:

A student shall be able to,

- understand the types of secondary steel making methods
- understood the advantages of decarburization, desulphurization, deoxidation
- aware about vacuum degassing methods of steel production
- make the steel of desired chemistry and cleanliness by performing the suitable treatments in "Ladle"
- understand the problems of steel making

Text books:

- Ahindra Ghosh, Secondary Steel Making; Principles and applications, 2000
- Gosh and A. Chatterjee;,Principles and Practices in Iron and Steel making, 2000

References:

- Dipak Mazumdar, A first course in Iron and Steel making, 2000
- Dutta S.K., Lele A.B., Metallurgical Thermodynamics Kinetics and Numericals ,2000

Category: Program Elective Course Subject code:MM3211

Electronic and Magnetic Materials

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- To know the electronic principles of magnetics, semiconductors, and superconductors
- To get knowledge on permanent magnets
- To describe the structure and processing of semiconducting materials
- To get knowledge on solar cells
- To discuss the structure and synthesis of superconducting materials

Unit-1: Magnetic Materials

Brief introduction: Origin of magnetism, Permanent magnetic moments of atoms, Magnetic domains, Superparamagnetism, Types of magnetic materials, Soft and hard magnetic materials; Ferritesclassification and crystal structure, Nanocrystalline soft magnetic materials, Melt-quenching method for soft magnetic ribbons and metallic glasses.

Unit-2: Permanent magnets

Properties and preparation of sintered magnets, high performance rare-earth permanent magnets (NdFeB & SmCo).

Unit-3: Semiconducting Materials

Brief introduction to semiconductor principles: band diagrams, doping, carrier concentration, I-V characteristics; Manufacturing of metallurgical grade silicon, Siemens process for electronic grade Si, Growth of single crystals by Czochralski process, Silicon wafer preparation for electronic devices. Techniques involved in preparation of electronic chips: Lithography, physical and chemical vapour deposition.

Unit-4: Solar cells

Photovoltaic effect, Types of silicon solar cells (singlecrystal, polycrystalline, & amorphous), Fabrication and functioning of silicon solar cells, Multi-junction solar cells.

Unit-5: Superconducting Materials

Temperature dependence of resistance, BCS theory, Type I and II superconductors, Meissner-Effect, High temperature Superconductors, YBCO & Bi-2212 – Crystal structure, Applications, Persistent current, Levitation; Synthesis of YBCO superconductor.

Course outcomes: The student will be able to

- Explain the electronic principles of magnetics, semiconductors, and superconductors
- Design microstructures for enhanced magnetic properties of materials
- Describe the structure and industrial processing of semiconducting materials
- Describe the structure and processing of solar cells.
- Discuss the structure and synthesis of superconducting materials

Text Books:

- B. D. Cullity, C.D. Graham, Introduction to Magnetic Materials, Wiley and IEEE, 2009
- J D Patterson, Solid State Physics, Springer, 2011.
- D.M. Hoffman, B. Singh, J.H. Thomas, Handbook of Vacuum Science and Technology, Academic Press, 1998.

Reference Books:

- John J. Croat, Rapidly Solidified Neodymium-Iron-Boron Permanent Magnets, Woodhead Publishing, 2017
- G.S. May and C.J. Spanos, Fundamentals of Semiconductor Manufacturing and Process Control, Wiley – IEEE, 2006.
- David Jiles, Introduction to Magnetism & Magnetic materials, London: Chapman & Hall, 1998.
- Milton Ohring, Materials Science of Thin Films, Academic Press, San Diego, 2002.

Category: Program Elective Course Subject code:MM4142

X-ray Powder Diffraction

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course Objectives:

- To get knowledge on geometry of crystals
- To enhance knowledge on XRD principles
- To analyse quantitative and qualitative measurements using XRD
- To understand the chemical analysis of materials.
- To understand the orientation behavior of crystalline materials

Unit-1: Geometry of crystals

Geometry of Crystals, Reciprocal Lattice, Stereographic Projection, Point Groups and Space Groups, Point Groups and Space Groups (Cont'd), Basics of X-Rays, Production and Detection of X-Rays.

Unit-2: Principles of X-ray Diffraction

Principles of X-Ray Diffraction, X-Ray Diffraction Methods, Debye Sherrer Camera, Diffractometer Measurements, Intensity of Diffracted Beams, Determination of Crystal structures, Precise Lattice Parameter Determination, XRD - Lab Demonstration.

Unit-3: Quantitative & Qualitative analysis

Phase Diagram Determination, Ordered Disordered Transformation, Qualitative Phase Analysis, Quantitative Phase Analysis 1, Precise Lattice Parameter Determination.

Unit-4: Chemical Analysis

Chemical Analysis by X-Ray Flourescence, Chemical Analysis by X-Ray Absorption, Effect of Crystallite Size on Diffracted X-Ray Intensity.

Unit-5: Texture and Residual stress

Texture Determination by XRD, Particle Size Determination by XRD, Effect of Crystallite Size on Diffracted X-Ray Intensity, Determination of Single Crystal Orientation by X-Rays, Stress Analysis by X-Rays, Factors Contributing to Peak Broadening, Residual Stress Measurement by X-Rays.

Course Outcomes: The student will be able to

- understand geometry of crystals
- gain knowledge on XRD principles
- analyse quantitative and qualitative measurements using XRD
- understand the chemical analysis of materials.
- understand the orientation behavior of crystalline materials

Textbooks:

- George.H.Stout and Lyle.H.Jensen, X-ray structure determination, Wiley, 1989
- Gregory S. Girolami, X-Ray Crystallography, University Science Books,U.S, 2015
- William Clegg, X-Ray Crystallography, OUP Oxford, 2015.

References:

- B.D.Cullity and S.R.Stock, Elements of X-Ray Diffraction, Prentice Hall, NJ, 2001.
- C. Suryanarayana, X-Ray diffraction, Springer, 1998
- Myeongkyu Lee, X-Ray Diffraction for Materials Research: From Fundamentals to Applications, Apple Academic Press, 2016.
- Yoshio Waseda, X-Ray Diffraction Crystallography: Introduction, Examples and Solved Problems, Springer, 2011.

Category: Program Elective Course Subject code: MM4141

Advanced Materials Processing

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course objectives:

- Explain the principle concepts of Smart materials, structures, Fibre optics, Magneto/Electro resistive Fluids, Biomimetics and MEMS with principles of working.
- Structure, processing, properties of smart materials will be included with type of activation of smart materials included – Thermal, mechanical, electrical, magnetic, chemical and optical means.
- Fundamental aspects in design and their integration of smart materials to wide range of applications.

Unit-1: Introduction

Introduction and classication of structural and functional materials; High T emperature Materials: Structure, Processing, mechanical behaviour and oxidation resistance of Stainless Steels, Ni- and Co- Based Superalloys, Aluminides and Silicides, Carbon-Carbon and Ceramic Composites;

Unit-2: Shape-Memory Alloys

Mechanisms of One-way and Two-way Shape Memory E-ect, Reverse Transformation, Thermoelasticity and Psuedoelasticity, Examples and Applications; Bulk Metallic Glass: Criteria for glass formation and stability, Examples and mechanical behaviour;

Unit-3: Nano-materials

Classication, size effect on structural and functional properties, Processing and properties of nanocrystalline materials, thin lms and multilayered coatings, single walled and multiwalled carbon nanotubes;

Unit-4: Magnetic Materials

Soft and hard magnetic materials for storage devices: Design and Processing; Piezoelectric Materials: Processing and Properties;

Unit-5: Advance processes

Advanced Processes applied for Advanced Materials: Single Crystal Growth, Rapid Solidication, Inert Gas Condensation, Physical and Chemical Vapour Deposition of Thin Films

Course outcomes: At the end of the course, students should be able to

- Understand the key practical theory with the operation principles of smart materials, their manufacturing, properties and their applications
- Address the key challenges and obstacles with manufacturing of different smart materials
- Design and justify appropriate materials for specific application related to smart structures.

Category: Program Elective Course Subject code: MM4141

Smart Materials

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-1-0-4

Course objectives:

- Explain the principle concepts of smart materials, structures, fiber optics, magneto/electro resistive fluids, biomimetics and MEMS with principles of working.
- Structure, processing, properties of smart materials will be included with type of activation of smart materials included – thermal, mechanical, electrical, magnetic, chemical and optical means.
- Fundamental aspects in design and their integration of smart materials to wide range of applications.

Unit-1: Introduction

introduction to smart materials : overview of the different types of smart materials, smart materials used in structures, smart material for sensors, actuators controls, memory and energy storage and their inter-relationships, concept of high bandwidth-low strain generating materials (HBLS), and low bandwidth high strain generating materials (LBHS), nano composite materials.

Unit-2: Electrically Activated Materials

Piezo – Piezoelectricity, Piezoresistivity; Ferroelectricity; Dielectrics, Dielectric Materials - Electrostrictive Materials - Electrets - Electrorheological Fluids. Ionic - Conductive Polymers - Ionomeric Polymer-Metal Composites - Carbon Nanotubes.

Unit-3: Thermally, Magnetic, chemical and optical Activated Materials

Shape memory alloys - Classification - Transformation - Ni-Ti Alloys, Shape memory ceramics - Shape memory polymers – Applications. Magnetostriction - Magnetorheological Fluid – Superconductors. Chemical Gels- Self healing materials. Optically activated polymers- Azobenzene- Liquid Crystals.

Unit-4: Smart Materials for Energy Applications

Materials used for energy storage, Hydrogen storage materials, energy harvesting, energy scavenging from vibrations.

Unit-5: Manufacturing Techniques for Smart Materials

Micro manufacturing, high resolution lithography, LIGA process, Generative manufacturing processes and Ion beam processes.

Course outcomes: At the end of the course, students should be able to

- understand the key practical theory with the operation principles of smart materials, their manufacturing, properties and their applications
- address the key challenges and obstacles with manufacturing of different smart materials
- design and justify appropriate materials for specific application related to smart structures.

Text books and references:

- Gandi, M.V. and Thompson, B.S., "Smart Materials and Structures," Chapman & Hall, UK, 1992
- B. Culshaw, "Smart Structures and Materials," Artech House, Inc., Norwood, USA, 1996.
- D. Leo, "Smart Materials Systems", Wiley 2007.
- K. Otsuka and C. M. Wayman, "Shape Memory Materials", Cambridge Press 1999.
- Brian Culshaw, Smart Structures and Materials, Artech House, 2000
- Gauenzi, P., Smart Structures, Wiley, 2009
- A.V.Srinivasan, 'Smart Structures –Analysis and Design", Cambridge University Press, New York, 2001.

Category: Seminar Internship Project	Subject code: MM3901

Project-1

Externals: 40 Marks	L-T-P-C*
Internals: 60 Marks	0-0-0-1

Objectives:

- To complete literature survey
- To identify the tools required for the research topic chosen
- To check the feasibility of completion of the project with the available facilities
- To find the alternative methods to conduct the experiments based on requirement
- To make a research plan for the chosen topic

Outcomes:

At the end of semester the student should be able to

- find the work carried out till dated
- find the gaps in the research work for the chosen area
- propose the objective of the project work
- find the tools required to carry out the proposed work
- developed a plan for the project work

Category: Seminar Internship Project Subject code: MM4901

Project-2

Externals: 40 Marks	L-T-P-C*
Internals: 60 Marks	0-0-0-4

Objectives:

- To carry systematic study of the research problem chosen
- To conduct the experiments or simulations for the chosen variables
- To carry systematic analysis of the results obtained
- To draw conclusions based on the results obtained
- To indentify the gaps in the research topic based on work carried and the results obtained

Outcomes:

At the end of semester the student should be able to

- conduct the experiments using the available facilities
- complete error analysis and validate the quality of the experiments and results
- present the results/work using suitable methods
- report the scope for continuation of work
- submit a well structured report

Category: Seminar Internship Project Subject code: MM4902

Project-3

Externals: 40 Marks L-T-P-C*
Internals: 60 Marks 0-0-0-6

Objectives:

- To carry systematic study of the research problem chosen
- To conduct the experiments or simulations for the chosen variables
- To carry systematic analysis of the results obtained
- To draw conclusions based on the results obtained
- To identify the gaps in the research topic based on work carried and the results obtained

Outcomes:

At the end of semester the student should be able to

- complete the planned experiments using the available facilities
- complete error analysis and validate the quality of the experimental results
- present the concluding remarks based on the results reported
- submit a well structured report

Category: Seminar Internship Project Subject code: MM4900

Summer Internship

Externals: 30 Marks L-T-P-C*

Internals: 70 Marks 0-0-0-1

Objectives:

- To understand the literature survey
- To get exposure of industry
- To get exposure of research lab
- To make them ready for project work of the chosen topic

Outcomes:

At the end of semester the student should be able to

- present the work carried out till dated
- present the literature survey
- prepare presentation slides
- communicate the work done

Category: Seminar Internship Project Subject code: MM4000

Comprehensive Viva

Externals: 30 Marks L-T-P-C*

Internals: 70 Marks 0-0-0-1

Objectives:

- To memorize the concepts of all subjects
- To get experience of technical interview
- To make them ready for competitive exams

Outcomes:

At the end of semester the student should be able to

- answer the concepts of core subjects well
- understood the technical interview
- ready to get through competitive exams
- communicate well

Category: Open Elective Course Subject code:MM4301

Non-destructive Testing

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-0-0-3

Course Objectives:

- To study and understand the basic principles of NDT methods
- To differentiate between NDT methods
- To study and understand NDT methods for detecting surface defects
- To study and understand NDT methods for detecting Internal defects
- To interpret defects in engineering components using NDT methods

Unit-1: Introduction

Introduction to NDT, Classification of NDT processes; advantages and limitations of NDT.

Eddy current testing and liquid penetrant testing: principle, advantages, limitations and applications of eddy current testing; liquid penetrant testing - Principle, Advantages, Limitations and applications.

Unit-2: Magnetic Particle Testing

Principle, procedure, magnetic media, methods to generate magnetic fields, method of de-magnetization, interpretation of indications, advantages, limitations and applications.

Unit-3: Radiographic Testing

Principle, gamma radiography, X-ray radiography and neutron radiography, imaging modalities, radiograph interpretation, precautions against radiation hazards.

Unit-4: Ultrasonic Testing and Acoustic Emission Testing

Principles of ultrasonic testing, ultrasonic propagation, test techniques – normal and angle beam, probes, transducers, applications advantages and limitations of ultrasonic testing.

Unit-5: Acoustic Emission Testing

Principle, sources of acoustic emission, acoustic emission signal parameters, industrial applications.

Course Outcomes:

The student should be able to

- comprehend the basic principles of NDT methods
- differentiate between NDT methods
- select NDT methods for detecting surface defects
- select NDT methods for detecting internal defects
- interpret defects in engineering components using NDT methods

Text Books:

1. Practical Non-Destructive Testing, Baldev Raj, T.Jayakumar, M.Thavasimuthu, Narosa Publishing House, 2014.

2. Non-Destructive Testing Techniques, Ravi Prakash, New Age International Publishers, 1st revised edition, 2010.

Reference Books:

1. Non-Destructive Testing - Theory, Practice and Industrial Applications, Wong B Stephen, LAP Lambert Academic Publishing, USA, 1st edition, 2014.

2. Non Destructive Test and Evaluation of Materials, J. Prasad and C. G. K. Nair, Tata McGraw-Hill Education, 2011, 2nd edition.

3. ASM Metals Handbook, Non-Destructive Evaluation and Quality Control, American Society of Metals, Metals Park, Ohio, USA, Volume-17.

4. Charles, J. Hellier, Handbook of Non-destructive evaluation, Charles, J. Hellier, McGraw Hill, New York, 2001.

5. Paul E Mix, Introduction to Non-destructive testing: a training guide, Wiley, 2 nd Edition New Jersey, 2005.

Online Resources:

1. www.ndt.edu

2. www.nde-ed.org

Category: Open Elective Course Subject code: MM4401

Materials Science

Externals: 60 Marks L-T-P-C*

Internals: 40 Marks 3-0-0-3

Course Objectives:

- Understand concepts on properties and selection of metals, ceramics, and polymers for design and manufacturing.
- Study variety of engineering applications through knowledge of atomic structure, electronic structure, chemical bonding, crystal structure, x-rays and x-ray diffraction, defect structure.
- Study Microstructure and structure-property relationships, Phase diagrams, heat treatment of steels.
- Study detailed information on types of corrosion and its prevention.
- Learn information on selection of materials for design and manufacturing.

Unit-1: Introduction

Engineering Materials – Classification – levels of structure. Crystal Geometry and Structure Determination, Space lattice and Unit cell. Bravais lattices, crystal systems with examples. Lattice coordinates, Miller indices, Bravais indices for directions and places: crystalline and non crystalline solids; ionic, covalent and metallic solids; packing efficiency, coordination number; structure determination by Bragg's X-ray diffraction and powder methods.

Unit-2: Crystal Imperfection

Point defects, line defects-edge and screw dislocation, Berger's circuit and Berger's vectors, dislocation reaction, dislocation motion, multiplication of dislocations during deformation , role of dislocation on crystal properties; surface defects, dislocation density and stress required to move dislocations.

Unit-3: Basic thermodynamic functions

Phase diagrams and phase transformation: Primary and binary systems-general types with examples; tie line& lever rule, non equilibrium cooling: phase diagrams of Fe-Fe_3-C, Pb-Sn, Cu-Ni systems. Phase transformations in Fe-Fe_3-C steels, Time-Temperature-Transformation (TTT) curves for eutectoid steels and plain carbon steels; effect of alloying elements on properties of steels; types of steels, alloys and other metals used in chemical industry.

Unit-4: Elastic and elastic and plastic deformations:

Elastic, an elastic and plastic deformations in solid materials; rubber like elasticity, visco elastic behavior (models); shear strength of real and perfect crystals, work hardening mechanisms, cold working, hot working; dynamic recovery, recrystallization, grain growth, grain size and yield stress, Brief description of heat treatment in steels.

Magnetic materials: Terminology and classification, magnetic moments due to electron spin, ferro-magnetism and related phenomena, domain structure, hysteresis loop, soft and hard magnetic materials.

Unit-5: Fracture in ductile and brittle materials, creep

Mechanism of creep and methods to reduce creeping in materials, creep rates and relations. Fatigue-mechanisms and methods to improve fatigue resistance in materials. Composite materials: types; stress-strain relations in composite materials, applications.

Oxidation and Corrosion: Mechanisms of oxidation, oxidation resistant materials, principles and types of corrosion, protection against corrosion.

Course Out Comes:

- Students will be able to understand the crystallographic changes in the materials which occur with the variation in temperature and apply the knowledge of structure property relationship in various metallurgical operations.
- An understanding of crystal geometry is developed and the student will be able to identify and index different planes and directions in metals and compounds which are of importance in metallurgical and chemical engineering operations.
- An insight into different defects in the crystals is given.
- An ability to predict solid solution formation of a given set of metals and associated phase transformations is developed by the study of phase diagrams.
- Structure property relations of several metals are analyzed with reference to their crystal structures, phase compositions, working conditions, defect concentration etc.,

Text Book:

1. Materials Science and Engineering, 5thed. V. Raghavan, PHI Learning Pvt. Ltd., New Delhi, 2009.
2. Physical Metallurgy – V. Raghavan
3. Material Science and Engineering – Callister
4. Introduction to Physical Metallurgy – Avner
5. Material science and Metallurgy – V. D. Kodgire, S. V. Kodgiri

Reference books:

1. Elements of Materials Science, L.R. Van Vlack,
2. Science of Engineering Materials, vols. 1&2, Manas Chanda, McMillan Company of India Ltd.

Category: Open Elective Course Subject code: MM4402

Nano Technology

Externals: 60 Marks L-T-P-C*
Internals: 40 Marks 3-0-0-3

Course Objectives:

- To Identify the need for nano materials.
- To Explain the bottom up and top-down approaches of nano material synthesis.
- To Describe the size effect on optical, electrical, mechanical, magnetic and thermal properties.
- To Review the applications of nano materials and nano devices
- To Understand the natural available nanomaterials

Unit-1: Classification of Nanomaterials

Introduction: defining nanotechnology, nanoscale comparison, Challenges of this size scale. Nanoscience vs Nanotechnology. Classification of Nanomaterials: Challenges of this Size Scale, Application area & opportunity, Classification According to their origin, dimensions & structural configuration; nanocomposite. Size Effect: Structural differences on the nanoscale, Factors Influencing Properties of nanomaterials, Surface Area and Energy, effect of size on optical, electrical, physical & chemical properties.

Unit-2: Nanocarbon

Classifications of carbon-based nanomaterials; carbon black & its applications; Fullerene: Definition, discovery and structure; Properties- Solubility, Hydrated Fullerene, Superconductivity, Endohedral fullerenes, Existence in the nature, Synthesis of C60. Carbon Nanotubes (CNT): Introduction, discovery, Rolling-up a graphene sheet to form a tube, Chiral Vector; Types of CNTs, Properties of CNTs, Synthesis of CNTs, Purification & Sorting of CNTs: The most common impurities, process of purification, Growth Mechanism of CNT: Tip-growth model & Base-growth model, Defects in Carbon nanotubes- Types of defects, Stone-Wales Defects, Role of defects on different properties. Graphene: Introduction, Graphene Superlatives, Discovery of Graphene, Salient features of the structure, Amazing Mechanical Properties, Electrical Properties, Thermal Conductivity, Optical properties, Synthesis of Graphene.

Unit-3: Applications of Nanomaterials

Early Applications, Applications in Electrical engineering, Electronics engineering, Mechanical engineering, Biodevices, Energy storage; New Applications-Glass-based electronics, Antibiotics, Ultrasonic Microphones, Lubricant, Mechanical Watch, Solar cells, Camera lenses, NFC Antenna, Super-Silk, Microbats, Water Purification, Sensors; Graphene's future, Challenges Surrounding Graphene.

Unit-4: Synthesis of nanomaterials

Physical Methods: Mechanical Methods, Methods based on Evaporation, Sputter Deposition, Chemical Vapor Deposition, Electric Arc Deposition, Ion Beam Technique, Molecular Beam Epitaxy. Chemical Methods: Synthesis of Metal nanoparticles by colloidal route, Sol-Gel Method, Hydrothermal synthesis, Sonochemical Synthesis, Microwave Synthesis. Nanofluids: Introduction, Characteristic, Methods for Producing Nanoparticles/Nanofluids, Nanofluid Structure, Effect of nanoparticles in NF, Advantages and challenges, Applications.

Ferrofluid: Introduction, Synthesis of Magnetite Nanocrystals, Applications.

Unit-5: Natural Nanomaterials

Introduction, Biomimicry, lotus effect and its applications, Phenomenon of iridescence in butterflies and its applications, Gecko's Sticky Feet, Shark Skin, Toucan Beaks, Water Striders.

Course outcomes: The student will be able to

- Identify the need for nano materials.
- Explain the bottom up and top-down approaches of nano material synthesis.
- Describe the size effect on optical, electrical, mechanical, magnetic and thermal properties.
- Review the applications of nano materials and nano devices
- Understand the natural available nanomaterials

Text Books:

- B. S. Murty et.al., Textbook of Nanoscience and Nanotechnology, Universities Press (India) Private Limited, 2013.
- Sulabha K. Kulkarni, Nanotechnology Principles and Practices, Capital Publishing Company, 2007.
- H. Hosono, Y. Mishima, H. Takezoe, K.J.D Mackenzie, Nanomaterials- From Research to Applications, Elsevier, 2008.

Reference:

- Massimilano Di Ventra, S. Evoy, James R. Heflin Jr, Introduction to Nanoscale Science and Technology, Springer, 2009.
- Charles P. Poole Jr., Frank J. Owens, Introduction to Nanotechnology, Wiley India, New Delhi, 2010.
- Jack Uldrich, Deb Newberry, Next Big Thing Is Really Small: How Nanotechnology Will Change the Future of Your Business, Crown Business Publications, 2003.